Beam-Structured DL Precoder Design for Massive MIMO Multiple LEO Satellite Communications

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***Abstract*—In this paper, we investigate the downlink (DL) precoder design for massive multiple-input multiple-output (MIMO) multiple low earth orbit (LEO) satellite communication (SATCOM). We first establish the DL beam based channel model for massive MIMO multiple LEO SATCOM systems by using sampled array response vectors. We propose a closed-form statistical channel state information (sCSI) based space domain DL precoder design for multi-satellite systems by maximizing the average signal-to-leakage-plus-noise ratio (ASLNR). Then, with the proposed beam based channel model, we transform the design of space domain DL precoder into that of lower-dimensional beam domain vector and the resulting space domain precoder is beam structured. Moreover, by leveraging the properties of the beam matrix, we propose a low-complexity design and implementation for the beam structured DL precoder. Simulation results demonstrate that the proposed beam structured DL precoder can achieve near performance to the space domain approach with significantly reduced computational complexity.**

***Index Terms*—Multiple LEO satellite communication, massive MIMO, beam structured DL precoder design, statistical CSI.**

I. Introduction

SATELLITE communication (SATCOM) is emerging as a critical technology for global connectivity, particularly in unserved and underserved regions where terrestrial networks face significant deployment challenges [1], [2], [3], [4], [5]. With advancements in satellite technology and the declining cost of space operations, low earth orbit (LEO) satellites are becoming an essential component of next-generation wireless communication systems [6], [7]. Unlike geostationary (GEO) satellites, which provide broad regional coverage but suffer from high latency, LEO satellites operate at altitudes between 200 to 2,000 km, enabling reduced propagation delay, lower power consumption, and improved service quality [8], [9], [10], [11]. These advantages make LEO constellations an integral part of future integrated space-terrestrial networks, complementing terrestrial 5G systems and supporting ubiquitous, high-throughput connectivity in remote areas.

Massive multiple-input multiple-output (MIMO) has played an essential role in current 5G systems and will serve as a foundation for the upcoming 6G networks [12], [13], [14], [15], [16], [17], [18], [19], [20]. Massive MIMO technology is extended to SATCOM to enhance connectivity, reliability, and data rates by providing improved spectral efficiency and supporting a large number of simultaneous user terminals (UTs) [21], [22], [23], [24], [25], [26]. A transmission framework for massive MIMO single satellite communication systems is introduced in [21], where the authors utilize statistical channel state information (sCSI) to design the uplink (UL) receiver and downlink (DL) precoder. Unlike instantaneous CSI (iCSI), which is difficult to acquire due to rapid variations caused by Doppler effects, sCSI remains stable over a longer period. Inspired by [21], several works have focused on the investigation of LEO SATCOM system with massive MIMO, including channel estimation [22], UL transmission [23], and DL transmission [24], [25], [26]. In [22], the authors propose a channel estimation method for massive MIMO SATCOM by exploiting the specific property of satellite channels. The authors of [23] investigate the UL transmit design to approach the ergodic sum rate capacity in massive MIMO SATCOM systems. Single data stream transmission strategy is proven to be optimal for sum rate maximization in [24], in which the DL precoding vector is designed to maximize the sum rate. In [25], the authors investigate the design of hybrid precoders in massive MIMO LEO SATCOM systems to maximize the energy efficiency. The secure transmission for SATCOM in both GEO and LEO scenarios is investigated in [26], where the robust secure transmission design is developed by using imperfect CSI. Note that the aforementioned works do not consider multiple satellite scenario.

The deployment of mega-constellations, comprising thousands of satellites, has emerged as a promising approach to enhancing spectral efficiency in SATCOM systems [27], [28], [29], [30], [31], [32]. Notably, SpaceX's Starlink project aims to deploy over 40,000 LEO satellites to provide global high speed internet coverage. Until now, more than 6,000 Starlink satellites have been launched into orbit, providing communication services to designed ground UTs [33], [34], [35], [36]. With more satellites in orbit, there may exist

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multiple satellites that are simultaneously visible to a single UT, which enables cooperative multi-stream transmissions from multiple satellites to a single UT, substantially improving data throughput. In [37], the authors investigate the beamspace MIMO for multi-satellite systems, where multiple satellites provide service to a single ground station, and the DL precoder and receiver are designed with their position information. For multi-satellite cooperative systems, the authors of [38] propose a low-complexity joint user scheduling and hybrid beamforming design method to improve the system spectral efficiency. In [39], the authors propose a metasurface based hybrid beamforming design for multi-LEO satellite systems with satellites at different orbital altitudes, accounting for hardware impairments. In [40], a deep machine learning based joint design of channel prediction and precoding is proposed for multibeam LEO satellite Internet of Things. In [41], the authors develop an analytical framework to evaluate UL/DL coverage and average delay in satellite-assisted mobile edge computing systems by using stochastic geometry and queuing theory. In [42], the authors investigate asynchronous interference mitigation in cooperative multi-LEO satellite systems, where the satellites jointly serve multiple single-antenna UTs. Each satellite applies a time-advance mechanism for each UT so that the desired signals arrive synchronously at the target UT, whereas signals intended for other UTs remain asynchronous at that UT. Moreover, the authors propose a cooperative beamforming algorithm with sCSI to maximize the achievable rate by considering the asynchronization.

For massive MIMO orthogonal frequency division multiplexing (OFDM) multi-satellite systems, time and frequency synchronizations are critical factors, as any misalignment can severely impact cooperative performance. Recently, several works have focused on the challenges posed by time and frequency asynchronizations, proposing techniques to address the effects of asynchronizations in both DL [43] and UL [44] scenarios for massive MIMO multiple LEO SATCOM. To mitigate the effects of frequency and time asynchronizations between satellites and UTs, a spatial receive processing method with frequency and time compensations at each UT is proposed in [43], where DL signal and channel models for OFDM are developed with full consideration of time and frequency asynchronizations, and the optimality of single data stream transmission for multi-satellite systems is confirmed. Moreover, a joint design of precoding and receiving vectors is proposed by utilizing sCSI only. In [44], the effects of time and frequency asynchronizations on signals received by each satellite, which are intended for different satellites and transmitted from ground UTs, is analyzed. Linear precoding with time and frequency pre-compensations is proposed at each UT for each satellite to mitigate the effects of asynchronizations. In addition, a matrix inversion free approach is proposed to jointly design the precoding and receiving vectors.

In this paper, by leveraging the spatial single path property of the satellite channel, we propose a low-complexity beam structured precoder design for massive MIMO multiple LEO SATCOM that accounts for time and frequency asynchronizations. Our major contributions are summarized as follows.

- We establish a DL beam based channel model (BBCM) for massive MIMO SATCOM by leveraging the physical characteristics of satellite channels. The proposed BBCM incorporates sampled array response vectors in the space domain, each corresponding to a physical spatial beam directed towards the sampled directional cosines.
- We propose a closed-form sCSI based space domain DL precoder for massive MIMO multi-satellite systems, aimed at maximizing the average signal-to-leakage-plus-noise ratio (ASLNR). Then we show that the design of space domain DL precoder can be transformed into that of lower-dimensional beam domain vector and the resulting space domain DL precoder is beam structured. Further, a beam selection criterion for the beam structured DL precoder design is provided.
- We present a low-complexity design and implementation for our proposed beam structured DL precoder by leveraging the properties of the beam matrix. Specifically, the lower-dimensional beam domain precoder can be computed by using only real-valued matrices, and the complexity can be significantly reduced.
- Simulation results demonstrate that the proposed beam structured DL precoder in massive MIMO multi-satellite communication systems can achieve near performance to the space domain approach, while significantly reducing the computational complexity.

The remaining sections are organized as follows. Section II presents the system model for massive MIMO multi-satellite system including the system configuration, the DL BBCM and signal models. Section III proposes the space domain DL precoder design with sCSI only, and the beam structured DL precoder design for massive MIMO multi-satellite system. Section IV presents the low-complexity design and implementation for the proposed beam structured DL precoder. Simulation results are provided in Section V. Section VI concludes this paper. The main symbols used in this paper are listed in Table I for ease of reference.

Notations: Throughout this paper, scalars, vectors, and matrices are represented by lowercase letters, bold lowercase letters, and bold uppercase letters, respectively. We denote $\mathbb{R}^{M\times N}$ ($\mathbb{C}^{M\times N}$) as the set of all real (complex) matrices of size $M \times N$. The transpose, conjugate, and conjugate transpose of a matrix or vector are denoted by $(\cdot)^T$, $(\cdot)^*$, and $(\cdot)^H$, respectively. We use $[\mathbf{g}]_{\mathcal{A}}$ to denote the sub-vector of the elements with the index set $\mathcal{A}$ of $\mathbf{g}$. The identity matrix of size N is denoted by $\mathbf{I}_N$. The expectation operator is represented by $\mathbb{E}\{\cdot\}$. We use $[\mathbf{N}]_{i,j}$ and $[\mathbf{n}]_i$ to refer to the (i,j)-th element of a matrix $\mathbf{N}$ and the i-th element of a vector $\mathbf{n}$, respectively. The Kronecker product is denoted by $\otimes$, while the Hadamard (element-wise) product is represented by $\odot$. The cardinality of a set $\mathcal{S}$ is written as $|\mathcal{S}|$, and the set difference between two sets $\mathcal{A}$ and $\mathcal{B}$ is represented by $\mathcal{A} \setminus \mathcal{B}$. The Euclidean norm of a vector $\mathbf{b}$ is represented by $\|\mathbf{b}\|$. The notation $\mathcal{CN}(\mathbf{0}, \mathbf{C})$ represents a circularly symmetric complex Gaussian distribution with zero mean and covariance matrix $\mathbf{C}$. The operator $\mathrm{vec}(\mathbf{A})$ denotes the vectorization of matrix $\mathbf{A}$ formed by stacking its columns into a single column

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TABLE I
SYMBOL LIST

Notation	Definition
M_{x}, M_{y}	Number of satellite antennas at x- and y- axes
$g^{\mathrm{x}}_{k_s,s}$, $g^{\mathrm{y}}_{k_s,s}$	Directional cosines
$\mathbf{g}_{k_s,s}$	DL array response vector
$\mathbf{h}_{k_s,s}$	DL channel vector in OFDM subcarrier
$\beta_{k_s,s}$	Channel power
F_{x}, F_{y}	Fine factors
N_{x}, N_{y}, N	Sampling numbers
$\mathbf{V}_{\mathrm{x}}$, $\mathbf{V}_{\mathrm{y}}$, $\mathbf{V}$	Beam matrices
$\mathcal{N}^{\mathrm{dia}}_{k_s,s}$, $\bar{\mathcal{N}}^{\mathrm{dia}}_{k_s,s}$	Beam index sets
$\mathbf{\Phi}_{k_s,s}$, $\bar{\mathbf{\Phi}}_{k_s,s}$	Beam selection matrices
$\tilde{\mathbf{g}}_{k_s,s}$, $\check{\mathbf{g}}_{k_s,s}$	Diffusion vectors
$\tilde{\mathbf{h}}_{k_s,s}$, $\check{\mathbf{h}}_{k_s,s}$	DL beam domain channel vectors
p_{k_s}	DL transmit power
$\mathbf{w}_{k_s}$	Normalized space domain precoder
$\bar{\mathbf{w}}_{k_s}$	Normalized beam domain precoder
$\eta^{\mathrm{slnr}}_{k_s}$, $\eta^{\mathrm{aslnr}}_{k_s}$, $\bar{\eta}^{\mathrm{aslnr}}_{k_s}$	power normalization factors
$\mathcal{R}^{\mathrm{sum}}_{\mathrm{dl}}$	DL ergodic sum rate

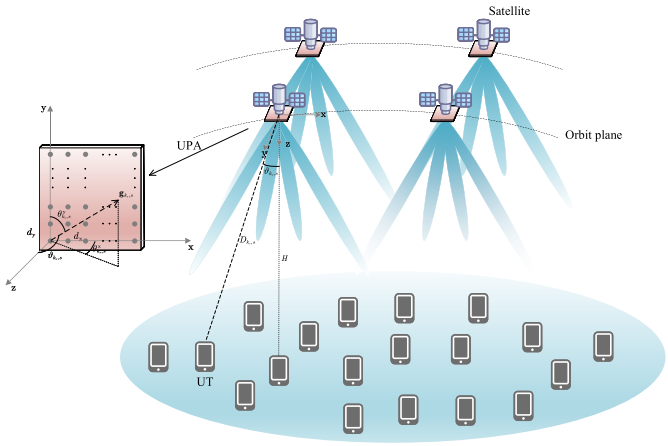


Fig. 1. Massive MIMO multiple LEO SATCOM system.

vector, and $\mathrm{uvec}(\mathbf{v})$ denotes the reverse operation of vec, reconstructing a matrix from its vectorized form $\mathbf{v}$.

II. System Model

In this section, we first present the system configuration of the multi-satellite massive MIMO-OFDM system. Then, we establish the DL BBCM by utilizing sampled array response vector and present the DL signal model.

A. System Configuration

Consider a multiple SATCOM system where S LEO satellites serve a number of single-antenna ground UTs simultaneously, as presented in Fig. 1. The satellite set is denoted as $\mathcal{S}$, with $|\mathcal{S}| = S$. Each satellite s, $s = 1, 2, \ldots, S$, provides services to K_s UTs, and is equipped with a UPA antenna incorporating M_{x} elements along the x-axis and M_{y} elements along the y-axis. At each satellite, the number of antenna elements is $M = M_{\mathrm{x}}M_{\mathrm{y}}$. Denote by k_s the k-th UT served by satellite s. The set $\mathcal{K}$ of all served UTs is denoted as $\mathcal{K} = \{k_s | k = 1, 2, \ldots, K_s, s = 1, 2, \ldots, S\}$, with $|\mathcal{K}| = K$. The set of served UTs of satellite s is denoted as $\mathcal{K}_s$ with $|\mathcal{K}_s| = K_s$. We assume that the satellite and UT positions are available with sufficient accuracy, which can be achieved via satellite ephemeris data and global navigation satellite system (GNSS) based UT localization.

OFDM modulation is employed in the wideband massive MIMO multiple SATCOM system with the number of cyclic prefix (CP) samples, N_{p}, and the number of subcarriers, N_{c}. Denote N_{t} as $N_{\mathrm{t}} = N_{\mathrm{p}} + N_{\mathrm{c}}$. The CP length is $T_{\mathrm{p}} = T_{\mathrm{s}}N_{\mathrm{p}}$, where T_{s} represents the system sampling period. The symbol duration excluding and including CP can be denoted as $T_{\mathrm{c}} = T_{\mathrm{s}}N_{\mathrm{c}}$ and $T_{\mathrm{t}} = T_{\mathrm{p}} + T_{\mathrm{c}}$, respectively.

B. DL BBCM

At each UT k_s, the Doppler shift and drifts caused by the motion of satellite s as well as the time delay and drifts due to the large propagation distance, are periodically compensated to track the time-varying Doppler and delay components resulting from the satellite trajectory. The residual Doppler and time delays become comparable to terrestrial systems, and the frequency and time for the link between satellite s and UT k_s can be regarded as synchronized, and the equivalent DL channel response from satellite s to UT k_s is expressed as[43]

$$\hat{\mathbf{h}}^{H}_{k_s,s}(n,m) = \sum_{\ell=0}^{P_{k_s,s}-1} \hat{g}_{k_s,s,\ell} e^{\jmath 2\pi \frac{n f^{\mathrm{ut}}_{k_s,s,\ell}}{N_{\mathrm{c}}}} \delta(m - m^{\mathrm{ut}}_{k_s,s,\ell}) \mathbf{g}^{H}_{k_s,s}, \tag{1}$$

where $P_{k_s,s}$ denotes the multipath number of the channel between UT k_s and satellite s, $\jmath \triangleq \sqrt{-1}$, and $\hat{g}_{k_s,s,\ell}$, $f^{\mathrm{ut}}_{k_s,s,\ell}$, and $m^{\mathrm{ut}}_{k_s,s,\ell}$ are, respectively, the effective channel gain, Doppler shift caused by the movement of UTs, and time delay due to the UTs' scattering environment, associated with path ℓ of the channel between UT k_s and satellite s. Moreover, $\mathbf{g}_{k_s,s}$ refers to the array response vector at satellite s for UT k_s, which is defined as

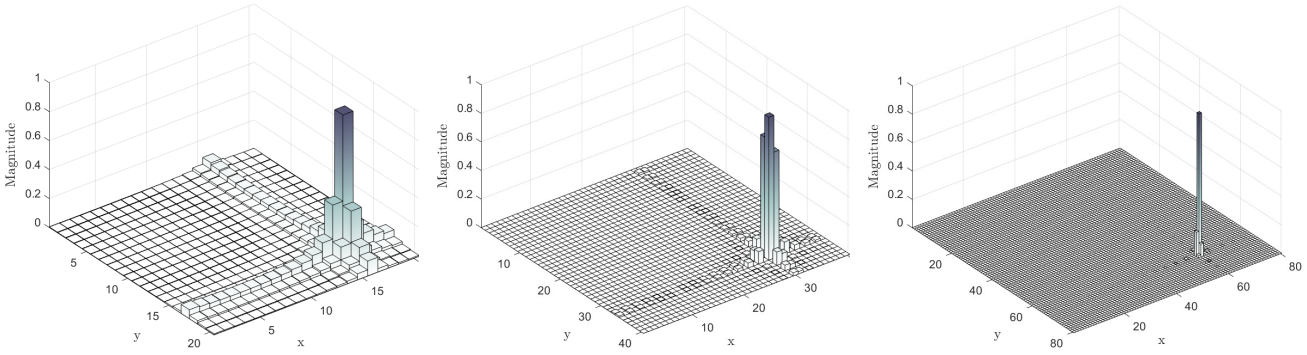
$$\mathbf{g}_{k_s,s} = \mathbf{g}^{\mathrm{x}}_{k_s,s} \otimes \mathbf{g}^{\mathrm{y}}_{k_s,s} = \mathbf{g}_{\mathrm{x}}(q^{\mathrm{x}}_{k_s,s}) \otimes \mathbf{g}_{\mathrm{y}}(q^{\mathrm{y}}_{k_s,s}), \tag{2}$$

with $\mathbf{g}_{\mathrm{v}}(q^{\mathrm{v}}_{k_s,s})$ expressed as

$$\mathbf{g}_{\mathrm{v}}(q^{\mathrm{v}}_{k_s,s}) = \frac{\omega_{\mathrm{v}}(q^{\mathrm{v}}_{k_s,s})}{\sqrt{M_{\mathrm{v}}}} \Big[1, e^{-\jmath 2\pi \frac{d_{\mathrm{v}}}{\lambda} q^{\mathrm{v}}_{k_s,s}}, \ldots, e^{-\jmath 2\pi (M_{\mathrm{v}}-1) \frac{d_{\mathrm{v}}}{\lambda} q^{\mathrm{v}}_{k_s,s}}\Big]^{T} \in \mathbb{C}^{M_{\mathrm{v}} \times 1}. \tag{3}$$

where $\omega_{\mathrm{v}}(q^{\mathrm{v}}_{k_s,s}) = e^{\jmath\pi(M_{\mathrm{v}}-1)\frac{d_{\mathrm{v}}}{\lambda} q^{\mathrm{v}}_{k_s,s}}$, $\mathrm{v} \in \{\mathrm{x}, \mathrm{y}\}$, λ denotes the carrier wavelength, d_{v} and M_{v} are, respectively, the antenna spacing and antenna number in the v-axis. $q^{\mathrm{x}}_{k_s,s} = \sin\theta^{\mathrm{y}}_{k_s,s} \cos\theta^{\mathrm{x}}_{k_s,s}$ and $q^{\mathrm{y}}_{k_s,s} = \cos\theta^{\mathrm{y}}_{k_s,s}$ are the directional cosines of satellite s for UT k_s along x-axis and y-axis, respectively, and $\theta^{\mathrm{x}}_{k_s,s}$ and $\theta^{\mathrm{y}}_{k_s,s}$ are the angles of satellite s for UT k_s along x-axis and y-axis, respectively. Note that $\omega_{\mathrm{v}}(q^{\mathrm{v}}_{k_s,s})$ is used to make $\mathbf{g}_{\mathrm{v}}(q^{\mathrm{v}}_{k_s,s})$ conjugate centrosymmetric, i.e. $[\mathbf{g}^{*}_{\mathrm{v}}(q^{\mathrm{v}}_{k_s,s})]_n = [\mathbf{g}_{\mathrm{v}}(q^{\mathrm{v}}_{k_s,s})]_{M_{\mathrm{v}}-n+1}$, which will be utilized to reduce the complexity of the

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(a) $F_{\mathrm{x}} = F_{\mathrm{y}} = 1$ (b) $F_{\mathrm{x}} = F_{\mathrm{y}} = 2$ (c) $F_{\mathrm{x}} = F_{\mathrm{y}} = 4$

Fig. 2. Magnitude of $|\tilde{\mathbf{g}}_{k_s,s}|$ for different fine factors.

subsequently proposed beam structured DL precoder design. The DL channel from satellite s to UT k_s in OFDM symbol p over subcarrier r can be expressed as

$$\mathbf{h}_{k_s,s,p,r}^{H} = \sum_{m=0}^{Q_{k_s,s}-1} \hat{\mathbf{h}}_{k_s,s}^{H}(pN_{\mathrm{t}}, m) e^{-j2\pi \frac{rm}{N_{\mathrm{c}}}} = \breve{g}_{k_s,s,p,r}^{*} \mathbf{g}_{k_s,s}^{H}, \quad (4)$$

where $Q_{k_s,s}$ is the length of $\hat{\mathbf{h}}_{k_s,s}(n,m)$, and $\breve{g}_{k_s,s,p,r}^{*} = \sum_{\ell=0}^{P_{k_s,s}-1} \hat{g}_{k_s,s,\ell} e^{j\frac{2\pi}{N_{\mathrm{c}}}\left(pN_{\mathrm{t}} f_{k_s,s,\ell}^{\mathrm{ut}} - r m_{k_s,s,\ell}^{\mathrm{ut}}\right)}$. Note that the satellite channel exhibits the spatial single path property as shown in (4). We assume that the channel gain has Rician distribution with the Rician factor $\kappa_{k_s,s}$, and the average channel power is $\mathbb{E}\{|\breve{g}_{k_s,s,p,r}|^2\} = \beta_{k_s,s}$. For simplicity, we ignore the subscripts for OFDM symbol and subcarrier, and denote by $\mathbf{h}_{k_s,s}$ the DL channel between satellite s and UT k_s on a specific subcarrier, which is given by

$$\mathbf{h}_{k_s,s} = \breve{g}_{k_s,s} \mathbf{g}_{k_s,s}. \quad (5)$$

Next, we derive a BBCM by uniformly sampling $q_{k_s,s}^{\mathrm{x}}$ and $q_{k_s,s}^{\mathrm{y}}$. Let $\mathcal{S}_{n_{\mathrm{v}}} = [2n_{\mathrm{v}}/N_{\mathrm{v}} - 1, 2(n_{\mathrm{v}}+1)/N_{\mathrm{v}} - 1)$, where $0 \le n_{\mathrm{v}} \le N_{\mathrm{v}} - 1$, and $N_{\mathrm{v}} = F_{\mathrm{v}} M_{\mathrm{v}}$ and F_{v} are, respectively, the sampling number and fine factor related to the v-axis. The directional cosines in $\mathcal{S}_{n_{\mathrm{v}}}$ can be approximated by $\xi_{n_{\mathrm{v}}}^{\mathrm{v}} = (2n_{\mathrm{v}} - N_{\mathrm{v}})/N_{\mathrm{v}}$. The beam matrix related to the v-axis is defined as

$$\mathbf{V}_{\mathrm{v}} = \left[\mathbf{g}_{\mathrm{v}}(\xi_0^{\mathrm{v}}), \mathbf{g}_{\mathrm{v}}(\xi_1^{\mathrm{v}}), \cdots, \mathbf{g}_{\mathrm{v}}(\xi_{N_{\mathrm{v}}-1}^{\mathrm{v}})\right]. \quad (6)$$

Note that each column of $\mathbf{V}_{\mathrm{v}}$ represents a specific beam in the space domain. When $N_{\mathrm{v}} = M_{\mathrm{v}}$, i.e., $F_{\mathrm{v}} = 1$, each column of $\mathbf{V}_{\mathrm{v}}$ is orthogonal to each other. We can reduce the sampling intervals in $\mathcal{S}_{n_{\mathrm{v}}}$ to obtain a higher resolution in angle by choosing $N_{\mathrm{v}} > M_{\mathrm{v}}$.

If the actual directional cosine of a UT, $q_{k_s,s}^{\mathrm{v}}$, exactly matches a sampled grid point, i.e., $q_{k_s,s}^{\mathrm{v}} = \xi_i^{\mathrm{v}}, \forall i = 0, 1, \ldots, N_{\mathrm{v}} - 1$, we can obtain

$$\mathbf{g}_{\mathrm{v}}(q_{k_s,s}^{\mathrm{v}}) = \mathbf{V}_{\mathrm{v}} \mathbf{e}_{i,N_{\mathrm{v}}}, \quad (7)$$

where $\mathbf{e}_{i,N_{\mathrm{v}}}$ represents a vector of length N_{v} with the i-th element being 1 and all other elements being 0, which indicates that all the energy is concentrated in the i-th beam. If a UT's actual directional cosine does not exactly correspond to any sampled value ξ_i^{v}, the associated response vector $\mathbf{g}_{\mathrm{v}}(q_{k_s,s}^{\mathrm{v}})$ cannot be directly mapped to a single column of $\mathbf{V}_{\mathrm{v}}$. In this case, $\mathbf{g}_{\mathrm{v}}(q_{k_s,s}^{\mathrm{v}})$ is given by

$$\mathbf{g}_{\mathrm{v}}(q_{k_s,s}^{\mathrm{v}}) = \mathbf{V}_{\mathrm{v}} \tilde{\mathbf{g}}_{\mathrm{v}}(q_{k_s,s}^{\mathrm{v}}), \quad (8)$$

where $\tilde{\mathbf{g}}_{\mathrm{v}}(q_{k_s,s}^{\mathrm{v}})$ is the diffusion vector that represents how the energy associated with an actual UT direction $q_{k_s,s}^{\mathrm{v}}$ is distributed across multiple nearby beams. For brevity, let $\tilde{\mathbf{g}}_{k_s,s}^{\mathrm{x}} = \tilde{\mathbf{g}}_{\mathrm{x}}(q_{k_s,s}^{\mathrm{x}})$ and $\tilde{\mathbf{g}}_{k_s,s}^{\mathrm{y}} = \tilde{\mathbf{g}}_{\mathrm{y}}(q_{k_s,s}^{\mathrm{y}})$. The DL satellite array response vector can be expressed as

$$\begin{aligned} \mathbf{g}_{k_s,s} &= \mathbf{g}_{k_s,s}^{\mathrm{x}} \otimes \mathbf{g}_{k_s,s}^{\mathrm{y}} \\ &= (\mathbf{V}_{\mathrm{x}} \otimes \mathbf{V}_{\mathrm{y}})(\tilde{\mathbf{g}}_{k_s,s}^{\mathrm{x}} \otimes \tilde{\mathbf{g}}_{k_s,s}^{\mathrm{y}}) = \mathbf{V} \tilde{\mathbf{g}}_{k_s,s}, \end{aligned} \quad (9)$$

where $\mathbf{V} = \mathbf{V}_{\mathrm{x}} \otimes \mathbf{V}_{\mathrm{y}} \in \mathbb{C}^{M \times N}$, $N = N_{\mathrm{x}} N_{\mathrm{y}}$, and $\tilde{\mathbf{g}}_{k_s,s} = \tilde{\mathbf{g}}_{k_s,s}^{\mathrm{x}} \otimes \tilde{\mathbf{g}}_{k_s,s}^{\mathrm{y}}$. Note that the beam matrix $\mathbf{V}$ is common for all UTs and independent from OFDM symbols and subcarriers. Let $\hat{\boldsymbol{\omega}}_{\mathrm{v}} = [1, e^{\jmath\pi}, \ldots, e^{\jmath\pi(M_{\mathrm{v}}-1)}]^T$ and $\boldsymbol{\omega}_{\mathrm{v}} = \frac{1}{\sqrt{M_{\mathrm{v}}}}[\omega_{\mathrm{v}}(\xi_0^{\mathrm{v}}), \omega_{\mathrm{v}}(\xi_1^{\mathrm{v}}), \ldots, \omega_{\mathrm{v}}(\xi_{N_{\mathrm{v}}-1}^{\mathrm{v}})]^T$. When $\frac{d_{\mathrm{v}}}{\lambda} = \frac{1}{2}$, $\mathbf{V}$ can be expressed as

$$\mathbf{V} = \mathrm{diag}\left(\hat{\boldsymbol{\omega}}\right) \left([\mathbf{F}_{N_{\mathrm{x}}}]_{1:M_{\mathrm{x}},:} \otimes [\mathbf{F}_{N_{\mathrm{y}}}]_{1:M_{\mathrm{y}},:} \right) \mathrm{diag}\left(\boldsymbol{\omega}\right), \quad (10)$$

where $\mathrm{diag}(\hat{\boldsymbol{\omega}}) = \mathrm{diag}(\hat{\boldsymbol{\omega}}_{\mathrm{x}}) \otimes \mathrm{diag}(\hat{\boldsymbol{\omega}}_{\mathrm{y}})$, $\mathbf{F}_{N_{\mathrm{v}}}$ is the N_{v}-point discrete Fourier transform (DFT) matrix, and $\mathrm{diag}(\boldsymbol{\omega}) = \mathrm{diag}(\boldsymbol{\omega}_{\mathrm{x}}) \otimes \mathrm{diag}(\boldsymbol{\omega}_{\mathrm{y}})$.

Due to the finite number of satellite antennas, the channel energy of UTs will diffuse among different beams. As illustrated in Fig. 2, when the fine factors equal to one, i.e., $F_{\mathrm{x}} = F_{\mathrm{y}} = 1$, the energy is distributed broadly across multiple beams, indicating significant leakage to neighboring beams. As F_{x} and F_{y} increase to 2, the leakage reduces, with more energy concentrated in fewer beams. For $F_{\mathrm{x}} = F_{\mathrm{y}} = 4$, the diffusion further diminishes, resulting in a sharper, more localized peak. This demonstrates that increasing the fine factor effectively suppresses energy leakage while confining the majority of the energy within the diamond-shaped region. Hereafter, we define the beam index set of chosen elements in $\tilde{\mathbf{g}}_{k_s,s}$ as

$$\begin{aligned} \mathcal{N}_{k_s,s}^{\mathrm{dia}} = \{(n_{\mathrm{x}}-1)N_{\mathrm{y}} + n_{\mathrm{y}} | n_{\mathrm{x}} = ((x_{\mathrm{idx}}^{\max} + \delta_{\mathrm{x}} - 1))_{N_{\mathrm{x}}} + 1, \\ n_{\mathrm{y}} = ((y_{\mathrm{idx}}^{\max} + \delta_{\mathrm{y}} - 1))_{N_{\mathrm{y}}} + 1, \end{aligned}$$

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$$x_{\text{idx}}^{\max} = \arg \max_{x_{\text{idx}} \in \{1,\cdots,N_{\text{x}}\}} [\tilde{\mathbf{g}}_{k_s,s}^{\text{x}}]_{x_{\text{idx}}},$$
$$y_{\text{idx}}^{\max} = \arg \max_{y_{\text{idx}} \in \{1,\cdots,N_{\text{y}}\}} [\tilde{\mathbf{g}}_{k_s,s}^{\text{y}}]_{y_{\text{idx}}},$$
$$\delta_{\text{x}}, \delta_{\text{y}} \in \{-\gamma, \cdots, \gamma\}, |\delta_{\text{x}}| + |\delta_{\text{y}}| \le \gamma\}, \tag{11}$$

where $|\mathcal{N}_{k_s,s}^{\text{dia}}| = N_{k_s,s}^{\text{dia}}$ and γ denotes the number of extra indices of the selected beams near the beam with the highest energy for x- and y- axes. Note that $\mathcal{N}_{k_s,s}^{\text{dia}}$ represents a beam index set corresponding to the beams arranged in a diamond shape, with the center located at a distance of γ from each of its four vertices. For each satellite, constructing the beam index set for each UT needs their relative positions. With GNSS providing meter-level horizontal accuracy for the UT [45] and modern precise orbit determination (POD) techniques achieving centimeter-level satellite-position errors [46], the resulting angular uncertainty is below the milliradian level when the satellite altitude is 600 km and will not affect the correctness of beam index sets.

Note that the requirement for continuous GNSS functionality on UTs, which is standard in modern devices, may increase energy consumption. Moreover, signaling overhead associated with transmitting precise UT location information may increase in high-mobility scenarios. Further, potential latency arising from GNSS acquisition can range from a few seconds to several minutes, depending on signal conditions and receiver sensitivity [47]. However, such latency mainly affects the initial positioning and has limited impact during steady-state operation, as subsequent updates can be performed incrementally [48].

With the beam index set $\mathcal{N}_{k_s,s}^{\text{dia}}$, $\mathbf{g}_{k_s,s}$ in (9) can be approximated as

$$\mathbf{g}_{k_s,s} \approx \check{\mathbf{V}}_{k_s,s} \check{\mathbf{g}}_{k_s,s}, \tag{12}$$

where $\check{\mathbf{V}}_{k_s,s} = \mathbf{V}\mathbf{\Phi}_{k_s,s}$, $\check{\mathbf{g}}_{k_s,s} = [\tilde{\mathbf{g}}_{k_s,s}]_{\mathcal{N}_{k_s,s}^{\text{dia}}}$, and $\mathbf{\Phi}_{k_s,s} \in \mathbb{R}^{N \times N_{k_s,s}^{\text{dia}}}$ is the beam selection matrix, defined as

$$[\mathbf{\Phi}_{k_s,s}]_{p,q} = \begin{cases} 1, & p = [\mathcal{N}_{k_s,s}^{\text{dia}}]_q, \\ 0, & \text{otherwise.} \end{cases} \tag{13}$$

Then, the DL channel in (5) can be expressed as

$$\mathbf{h}_{k_s,s} = \mathbf{V}\tilde{\mathbf{h}}_{k_s,s} \approx \check{\mathbf{V}}_{k_s,s} \check{\mathbf{h}}_{k_s,s}, \tag{14}$$

where $\tilde{\mathbf{h}}_{k_s,s} = \breve{g}_{k_s,s}\tilde{\mathbf{g}}_{k_s,s}$ denotes the DL channel in the beam domain, and $\check{\mathbf{h}}_{k_s,s} = \breve{g}_{k_s,s}\check{\mathbf{g}}_{k_s,s}$. We refer to this channel representation in (14) as BBCM.

Hereafter, we introduce additional N_{v}' sampling points $\{\xi_{i,j}^{\text{v}}\}_{j=0}^{N_{\text{v}}'-1}$ within $[\xi_i^{\text{v}}, \xi_{i+1}^{\text{v}})$ to obtain an easy-handling expression of $\mathbf{g}_{k_s,s}^{\text{v}}$. Let $\xi_{i,0}^{\text{v}} = \xi_i^{\text{v}}$. Denote by $\xi_{k_s,s}^{\text{v,min}}$ the point closest to $q_{k_s,s}^{\text{v}}$, which is defined as

$$\xi_{k_s,s}^{\text{v,min}} = \arg \min_{\xi_{i,j}^{\text{v}}} |\xi_{i,j}^{\text{v}} - q_{k_s,s}^{\text{v}}|. \tag{15}$$

Approximating the array response vector of $q_{k_s,s}^{\text{v}}$ in (8) by that of $\xi_{k_s,s}^{\text{v,min}}$ yields

$$\mathbf{g}_{k_s,s}^{\text{v}} = \mathbf{g}_{\text{v}}(q_{k_s,s}^{\text{v}}) \approx \mathbf{g}_{\text{v}}(\xi_{k_s,s}^{\text{v,min}}) = \mathbf{V}_{\text{v}}\tilde{\mathbf{g}}_{\text{v}}(\xi_{k_s,s}^{\text{v,min}}). \tag{16}$$

In this case, a sparse representation of the diffusion vector $\tilde{\mathbf{g}}_{\text{v}}(\xi_{k_s,s}^{\text{v,min}})$ can be obtained by using a sparse signal recovery algorithm, e.g. multiple measurement vector focal underdetermined system solver (MFOCUSS) [49]. Note that $\tilde{\mathbf{g}}_{\text{v}}(\xi_{i,j}^{\text{v}})$ can be obtained from $\tilde{\mathbf{g}}_{\text{v}}(\xi_{0,j}^{\text{v}})$ through a circular index shift:

$$\left[\tilde{\mathbf{g}}_{\text{v}}(\xi_{i,j}^{\text{v}})\right]_m = e^{\jmath\pi\frac{(M_{\text{v}}-1)i}{N_{\text{v}}}}([\boldsymbol{\omega}_{\text{v}}]_m)^{-1}\left[\text{diag}(\boldsymbol{\omega}_{\text{v}})\tilde{\mathbf{g}}_{\text{v}}(\xi_{0,j}^{\text{v}})\right]_{((m-i-1))_{N_{\text{v}}}+1}, \tag{17}$$

where $((m-i-1))_{N_{\text{v}}}$ denotes the modulo operation of N_{v}. The diffusion vectors $\tilde{\mathbf{g}}_{\text{v}}(\xi_{i,j}^{\text{v}}), \forall i,j$, in (17) constitute the diffusion vector codebook.

C. DL Signal Model

After proper frequency and time synchronizations, the received signal at UT k_s in the frequency domain within the OFDM framework can be given by [43]

$$y_{k_s} = \mathbf{h}_{k_s,s}^H \mathbf{x}_s + z_{k_s} = \mathbf{h}_{k_s,s}^H \sum_{i=1}^{K_s} \mathbf{x}_{i_s} + z_{k_s}, \tag{18}$$

where $\mathbf{x}_s$ and $\mathbf{x}_{i_s}$ are the transmit signal of satellite s, and the transmit signal of satellite s for UT i_s in the frequency domain, respectively. Here we also ignore the subscripts for OFDM symbol and subcarrier for simplicity.

Note that even though proper frequency and time synchronizations have been performed at UT k_s for satellite s, the signals from other satellites ($j \neq s$) typically remain asynchronous at UT k_s and are treated as Gaussian interference [43]. z_{k_s} in (18) denotes asynchronous interference signals from other satellites plus the noise signal at UT k_s, whose variance is given by

$$R_{k_s} = \mathbb{E}\{z_{k_s} z_{k_s}^H\} = \sum_{j=1, j\neq s}^{S} \beta_{k_s,j} \mathbb{E}\left\{\left|\mathbf{g}_{k_s,j}^H \mathbf{x}_j\right|^2\right\} + \sigma_{k_s}^2, \tag{19}$$

where $\sigma_{k_s}^2$ is the variance of the noise signal at UT k_s. The signal from satellite s to UT k_s will leak to UTs served by other satellites, and the variance of the asynchronous leakage signal is given by

$$\tilde{L}_{k_s} = \sum_{j=1, j\neq s}^{S} \sum_{i=1}^{K_j} \beta_{i_j,s} \mathbb{E}\left\{\left|\mathbf{g}_{i_j,s}^H \mathbf{x}_{k_s}\right|^2\right\}. \tag{20}$$

III. DL Precoder Design With sCSI

In this section, we first propose the design of space domain DL precoder in multiple LEO SATCOM systems by utilizing slow-varying sCSI. Then, with the established DL BBCM, we show that the design of space domain DL precoder can be transformed into that of the lower-dimensional beam domain vector, and the resulting space domain precoder is beam structured. In addition, we provide a beam selection criterion for the beam structured precoder design.

A. Space Domain Precoder Design

We consider that each satellite adopts linear precoding for generating the transmit signal, then (18) can be rewritten as

$$y_{k_s} = \mathbf{h}_{k_s,s}^H \sum_{i=1}^{K_s} \sqrt{p_{i_s}} \mathbf{w}_{i_s} s_{i_s} + z_{k_s}, \tag{21}$$

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where p_{k_s} is the allocated transmit power of satellite s for UT k_s, $\mathbf{w}_{k_s} \in \mathbb{C}^{M\times 1}$ denotes the normalized precoder of satellite s for UT k_s, with $\|\mathbf{w}_{k_s}\| = 1$, s_{k_s} denotes the transmit data stream of satellite s for UT k_s with zero mean and unit variance.

SLNR is an efficient and commonly used design metric for DL multi-cell multi-user MIMO systems [50]. We first consider the precoding approach based on the SLNR maximization criterion. Specifically, the SLNR of UT k_s can be expressed as

$$\mathrm{SLNR}_{k_s} = \frac{\left|\mathbf{h}_{k_s,s}^H \mathbf{w}_{k_s}\right|^2}{\sum_{i\neq k}^{K_s}\left|\mathbf{h}_{i_s,s}^H \mathbf{w}_{k_s}\right|^2 + L_{k_s} + \frac{\sigma_{k_s}^2}{p_{k_s}}}, \tag{22}$$

where $L_{k_s} = \sum_{j=1,j\neq s}^{S}\sum_{i=1}^{K_j}\beta_{ij,s}|\mathbf{g}_{ij,s}^H\mathbf{w}_{k_s}|^2$. The precoder that can maximize SLNR_{k_s} in (22) is calculated as

$$\mathbf{w}_{k_s}^{\mathrm{slnr}} = \frac{1}{\eta_{k_s}^{\mathrm{slnr}}}\left(\sum_{j\neq s}^{S}\sum_{i=1}^{K_j}\beta_{ij,s}\mathbf{g}_{ij,s}\mathbf{g}_{ij,s}^H + \sum_{i=1}^{K_s}\mathbf{h}_{i_s,s}\mathbf{h}_{i_s,s}^H + \frac{\sigma_{k_s}^2}{p_{k_s}}\mathbf{I}\right)^{-1}\mathbf{h}_{k_s,s}, \forall k_s, \tag{23}$$

where $\eta_{k_s}^{\mathrm{slnr}}$ denotes the power normalization factor that ensures $\|\mathbf{w}_{k_s}^{\mathrm{slnr}}\| = 1$. Note that the DL precoder that maximizes SLNR requires the iCSI of the channel $\mathbf{h}_{i_s,s}, \forall i \in \mathcal{K}_s, s \in \mathcal{S}$. However, it is generally infeasible for each satellite to obtain precise DL iCSI.

In the following, we propose the design of DL precoder for multiple LEO SATCOM by using only long-term sCSI at each satellite. We consider the ASLNR design metric, expressed as

$$\begin{aligned}\mathrm{ASLNR}_{k_s} &= \frac{\mathbb{E}\left\{\left|\mathbf{h}_{k_s,s}^H\mathbf{w}_{k_s}\right|^2\right\}}{\mathbb{E}\left\{\sum_{i\neq k}^{K_s}\left|\mathbf{h}_{i_s,s}^H\mathbf{w}_{k_s}\right|^2\right\} + L_{k_s} + \frac{\sigma_{k_s}^2}{p_{k_s}}} \\ &= \frac{\beta_{k_s,s}\left|\mathbf{g}_{k_s,s}^H\mathbf{w}_{k_s}\right|^2}{\sum_{i\neq k}^{K_s}\beta_{i_s,s}\left|\mathbf{g}_{i_s,s}^H\mathbf{w}_{k_s}\right|^2 + L_{k_s} + \frac{\sigma_{k_s}^2}{p_{k_s}}}.\end{aligned} \tag{24}$$

The DL precoder maximizing ASLNR_{k_s} in (24) can be calculated as

$$\mathbf{w}_{k_s}^{\mathrm{aslnr}} = \frac{1}{\eta_{k_s}^{\mathrm{aslnr}}}\left(\sum_{j=1}^{S}\sum_{i=1}^{K_j}\beta_{ij,s}\mathbf{g}_{ij,s}\mathbf{g}_{ij,s}^H + \frac{\sigma_{k_s}^2}{p_{k_s}}\mathbf{I}\right)^{-1}\mathbf{g}_{k_s,s}, \forall k_s, \tag{25}$$

where $\eta_{k_s}^{\mathrm{aslnr}}$ denotes the power normalization factor that ensures $\|\mathbf{w}_{k_s}^{\mathrm{aslnr}}\| = 1$. The corresponding maximum value of ASLNR can be given by

$$\mathrm{ASLNR}_{k_s}^{\max} = -1 + \frac{1}{1 - \beta_{k_s,s}\mathbf{g}_{k_s,s}^H\check{\mathbf{G}}_s^{-1}\mathbf{g}_{k_s,s}}, \tag{26}$$

where $\check{\mathbf{G}}_s = \sum_{j=1}^{S}\sum_{i=1}^{K_j}\beta_{ij,s}\mathbf{g}_{ij,s}\mathbf{g}_{ij,s}^H + \frac{\sigma_{k_s}^2}{p_{k_s}}\mathbf{I}$. Note that (25) provides a closed-form DL precoder based on sCSI that maximizes the ASLNR. The required sCSI includes the average channel power and the array response vectors, i.e., $\beta_{i_j,s}$ and $\mathbf{g}_{i_j,s}$, $\forall i_j$. Furthermore, based on the definition of the array response vector in (2), only the directional cosines, $q_{i_j,s}^{\mathrm{x}}$ and $q_{i_j,s}^{\mathrm{y}}$, are required to obtain $\mathbf{g}_{i_j,s}$. Notice that the sCSI based precoder design can be shared for different subcarriers and a number of OFDM symbols over a period that the sCSI keeps relatively stable, and therefore the complexity for the design of the DL precoder can be reduced significantly compared with the iCSI based approach.

B. Beam Structured Precoder Design

In this subsection, we propose the beam structured precoder design with the established BBCM. We first obtain the following theorem.

Theorem 1: When $M \to \infty$, and $\mathcal{N}_{k,s}^{\mathrm{dia}} \cap \mathcal{N}_{i,s}^{\mathrm{dia}} = \emptyset, \forall k, i, s$, the space domain DL precoder in (25) can be expressed as

$$\mathbf{w}_{k_s}^{\mathrm{aslnr}} = \check{\mathbf{V}}_{k_s,s}\check{\mathbf{w}}_{k_s}^{\mathrm{aslnr}}, \forall k_s, \tag{27}$$

where

$$\begin{aligned}\check{\mathbf{w}}_{k_s}^{\mathrm{aslnr}} = &\arg\max_{\check{\mathbf{w}}_{k_s}} \mathrm{ASLNR}_{k_s}^{\mathrm{b}}\left(\check{\mathbf{w}}_{k_s}, \mathcal{N}_{k_s,s}^{\mathrm{dia}}\right), \\ &\text{s.t. } \|\check{\mathbf{w}}_{k_s}\|^2 = 1, \forall k_s,\end{aligned} \tag{28}$$

with

$$\begin{aligned}&\mathrm{ASLNR}_{k_s}^{\mathrm{b}}\left(\check{\mathbf{w}}_{k_s}, \mathcal{N}_{k_s,s}^{\mathrm{dia}}\right) \\ &= \frac{\beta_{k_s,s}\left|\check{\mathbf{w}}_{k_s}^H\check{\mathbf{g}}_{k_s,s}\right|^2}{\sum_{i\neq k}^{K_s}\beta_{i_s,s}\left|\check{\mathbf{w}}_{k_s}^H\check{\mathbf{V}}_{k_s,s}^H\check{\mathbf{V}}_{i_s,s}\check{\mathbf{g}}_{i_s,s}\right|^2 + \check{L}_{k_s} + \frac{\sigma_{k_s}^2}{p_{k_s}}},\end{aligned} \tag{29}$$

and $\check{L}_{k_s} = \sum_{j\neq s}^{S}\sum_{i=1}^{K_j}\beta_{ij,s}|\check{\mathbf{g}}_{ij,s}^H\check{\mathbf{V}}_{ij,s}^H\check{\mathbf{V}}_{k_s,s}\check{\mathbf{w}}_{k_s}|^2$.

Proof: See Appendix A. ∎

From Theorem 1, when the satellite antenna number M is sufficiently large, and beam domain channels between the satellite and any two UTs are non-overlapping, the design of precoders $\mathbf{w}_{k_s} \in \mathbb{C}^{M\times 1}, \forall k_s$ can be transformed into that of beam domain vectors $\check{\mathbf{w}}_{k_s} \in \mathbb{C}^{N_{k_s,s}^{\mathrm{dia}}\times 1}, \forall k_s$. Due to the sparsity of beam domain channels in massive MIMO SATCOM systems, $N_{k_s,s}^{\mathrm{dia}}$ is typically much smaller than M, then the complexity for designing the DL precoder can be significantly reduced. We refer to $\check{\mathbf{w}}_{k_s}^{\mathrm{aslnr}}$ as the beam domain precoder and the resulting space domain precoder $\mathbf{w}_{k_s}^{\mathrm{aslnr}}$ as the beam structured precoder in the space domain, and $\mathrm{ASLNR}_{k_s}^{\mathrm{b}}(\check{\mathbf{w}}_{k_s}, \mathcal{N}_{k_s,s}^{\mathrm{dia}})$ in (29) refers to the ASLNR of the beam structured precoder in (27) with beam index set $\mathcal{N}_{k_s,s}^{\mathrm{dia}}$.

Note that overlapping beams between two UTs are identified by comparing their beam index sets (or equivalently, their diffusion vectors), obtained from the pre-computed codebook and directional cosines of UTs, as illustrated in (15)-(17). Because the codebook depends only on the sampled directional cosines, it can be generated offline and stored on-board. Hence, the complexity of identifying overlapping beams is negligible in the DL precoder design.

In practical systems, the beam domain channels of different UTs may overlap, which leads to performance degradation. Hence, additional beams need to be considered in the beam structured precoder design to enhance robustness and improve the system performance. Define a new beam index set $\bar{\mathcal{N}}_{k_s,s}^{\mathrm{dia}}$, whose definition is similar to that of $\mathcal{N}_{k_s,s}^{\mathrm{dia}}$ in (11), except that γ is replaced with $\bar{\gamma}$, and $\bar{\gamma} \geq \gamma + 1$. In the following theorem,

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we provide a beam selection criterion for the beam structured precoder design.

Theorem 2: When $M \to \infty$, we can obtain that

$$\mathrm{ASLNR}_{k_s}^{\mathrm{b}}\left(\check{\mathbf{w}}_{k_s}^{\mathrm{aslnr}}, \bar{\mathcal{N}}_{k_s,s}^{\mathrm{dia}}\right) \geq \mathrm{ASLNR}_{k_s}^{\mathrm{b}}\left(\check{\mathbf{w}}_{k_s}^{\mathrm{aslnr}}, \mathcal{N}_{k_s,s}^{\mathrm{dia}}\right), \tag{30}$$

where the equality holds when elements $\{\bar{n}_{k_s,s,m}\}_{m=1}^{\bar{N}_{k_s,s}^{\mathrm{dia}}-N_{k_s,s}^{\mathrm{dia}}}$ in the newly selected beam index set $\bar{\mathcal{N}}_{k_s,s}^{\mathrm{dia}} \setminus \mathcal{N}_{k_s,s}^{\mathrm{dia}}$ satisfy

$$\bar{n}_{k_s,s,m} \notin \bigcup_{i\in\mathcal{K}_j, j\in\mathcal{S}, \mathcal{N}_{i_j,s}^{\mathrm{dia}} \cap \mathcal{N}_{k_s,s}^{\mathrm{dia}} \neq \emptyset} \mathcal{N}_{i_j,s}^{\mathrm{dia}},$$
$$\forall s \in \mathcal{S}, k \in \mathcal{K}_s, m = 1, 2, \ldots, \bar{N}_{k_s,s}^{\mathrm{dia}} - N_{k_s,s}^{\mathrm{dia}}. \tag{31}$$

Proof: See Appendix B. ∎

Theorem 2 reveals that for the beam structured precoder design, using only the beams of UT k_s may not be optimal when its channel overlaps with those of other UTs in the beam domain. Therefore, additional beams should be incorporated into the beam structured precoder design. In addition, (31) is a criterion for selecting additional beams. Specifically, beams satisfying the condition in (31) should be excluded, as they will not contribute to improving the performance of the beam structured precoder. Moreover, the beams satisfying $\bar{n}_{k_s,s,m} \in \bigcup_{i\in\mathcal{K}_j, j\in\mathcal{S}, \mathcal{N}_{i_j,s}^{\mathrm{dia}} \cap \mathcal{N}_{k_s,s}^{\mathrm{dia}} \neq \emptyset} \mathcal{N}_{i_j,s}^{\mathrm{dia}}$ should be considered to improve the performance.

Even though the number of antennas at satellites is limited in practical systems, the beam structured precoder design as discussed above can still achieve nearly optimal performance with a large but finite number of satellite antennas. Hereafter, we restrict the space domain precoder for UT k_s to satisfy the following structure:

$$\mathbf{w}_{k_s} = \bar{\mathbf{V}}_{k_s,s}\bar{\mathbf{w}}_{k_s}, \tag{32}$$

where $\bar{\mathbf{V}}_{k_s,s} = \mathbf{V}\bar{\mathbf{\Phi}}_{k_s,s}$, and $\bar{\mathbf{\Phi}}_{k_s,s} \in \mathbb{R}^{N\times \bar{N}_{k_s,s}^{\mathrm{dia}}}$ is defined as

$$\left[\bar{\mathbf{\Phi}}_{k_s,s}\right]_{p,q} = \begin{cases} 1, & p = \left[\bar{\mathcal{N}}_{k_s,s}^{\mathrm{dia}}\right]_q, \\ 0, & \text{else.} \end{cases} \tag{33}$$

Therefore, the ASLNR can be written as in (34), as shown at the bottom of the page,where $\tilde{\mathbf{B}}_{k_s,i_j,s} = \bar{\mathbf{V}}_{k_s,s}^H \check{\mathbf{V}}_{i_j,s}$ and $\bar{\mathbf{B}}_{k_s,i_j,s} = \bar{\mathbf{V}}_{k_s,s}^H \bar{\mathbf{V}}_{i_j,s}$. The corresponding beam domain precoder that maximizes $\mathrm{ASLNR}_{k_s}^{\mathrm{b}}$ in (34) can be calculated as

$$\bar{\mathbf{w}}_{k_s}^{\mathrm{aslnr}} = \frac{1}{\bar{\eta}_{k_s}^{\mathrm{aslnr}}}\left(\sum_{j=1}^{S}\sum_{i=1}^{K_j}\beta_{i_j,s}\tilde{\mathbf{B}}_{k_s,i_j,s}\check{\mathbf{g}}_{i_j,s}\check{\mathbf{g}}_{i_j,s}^H\tilde{\mathbf{B}}_{k_s,i_j,s}^H + \frac{\sigma_{k_s}^2}{p_{k_s}}\bar{\mathbf{B}}_{k_s,k_s,s}\right)^{-1}\tilde{\mathbf{B}}_{k_s,k_s,s}\check{\mathbf{g}}_{k_s,s}, \forall k_s, \tag{35}$$

where $\bar{\eta}_{k_s}^{\mathrm{aslnr}}$ is chosen to make $(\bar{\mathbf{w}}_{k_s}^{\mathrm{aslnr}})^H\bar{\mathbf{B}}_{k_s,k_s,s}\bar{\mathbf{w}}_{k_s}^{\mathrm{aslnr}} = 1$. Then, the beam structured precoder is given by

$$\mathbf{w}_{k_s}^{\mathrm{aslnr}} = \bar{\mathbf{V}}_{k_s,s}\bar{\mathbf{w}}_{k_s}^{\mathrm{aslnr}}. \tag{36}$$

The corresponding maximum value of ASLNR is given by

$$\mathrm{ASLNR}_{k_s}^{\mathrm{b,max}} = -1 + \frac{1}{1 - \beta_{k_s,s}\check{\mathbf{g}}_{k_s,s}^H\tilde{\mathbf{B}}_{k_s,k_s,s}^H\left(\bar{\mathbf{L}}_{k_s}^{\mathrm{b}}\right)^{-1}\tilde{\mathbf{B}}_{k_s,k_s,s}\check{\mathbf{g}}_{k_s,s}}, \tag{37}$$

where $\bar{\mathbf{L}}_{k_s}^{\mathrm{b}} = \sum_{j=1}^{S}\sum_{i=1}^{K_j}\beta_{i_j,s}\tilde{\mathbf{B}}_{k_s,i_j,s}\check{\mathbf{g}}_{i_j,s}\check{\mathbf{g}}_{i_j,s}^H\tilde{\mathbf{B}}_{k_s,i_j,s}^H + \frac{\sigma_{k_s}^2}{p_{k_s}}\bar{\mathbf{B}}_{k_s,k_s,s}$. Note that the beams selected in $\bar{\mathcal{N}}_{k_s,s}^{\mathrm{dia}}$ can be adjusted flexibly to balance the complexity and performance, and (35) provides a closed-form expression for the sCSI based beam domain precoder whose dimension is typically much smaller than that of the space domain precoder in (25), thereby reducing the payload cost and complexity in SATCOM. The sCSI adopted in our designs refers to slowly varying parameters, i.e., directional cosines and channel powers. These parameters evolve on a much longer timescale than iCSI [21], [24], [43]. As the satellite networks continue to grow in density, the frequency of handovers between satellite will increase [51]. When a handover occurs, the newly serving satellite should recompute its DL precoder for the UT with the required sCSI, which allows low-overhead precoder updates during frequent satellite handovers.

IV. Low-Complexity Design and Implementation for the Beam Structured Precoder

In this section, we first show that the computation of the beam domain precoder relies solely on real-valued matrices, and present a low-complexity design and implementation for our proposed beam structured precoder by utilizing the properties of the beam matrix.

A. Low-Complexity Design

The design complexity of the beam structured precoder comes from (35), which can be rewritten as

$$\bar{\mathbf{w}}_{k_s}^{\mathrm{aslnr}} = \frac{1}{\bar{\eta}_{k_s}^{\mathrm{aslnr}}}\left(\sum_{j=1}^{S}\sum_{i=1}^{K_j}\beta_{i_j,s}\bar{\mathbf{\Phi}}_{k_s,s}^H\mathbf{V}^H\mathbf{V}\tilde{\mathbf{g}}_{i_j,s}\tilde{\mathbf{g}}_{i_j,s}^H\mathbf{V}^H\mathbf{V}\bar{\mathbf{\Phi}}_{k_s,s} + \frac{\sigma_{k_s}^2}{p_{k_s}}\bar{\mathbf{\Phi}}_{k_s,s}^H\mathbf{V}^H\mathbf{V}\bar{\mathbf{\Phi}}_{k_s,s}\right)^{-1}\bar{\mathbf{\Phi}}_{k_s,s}^H\mathbf{V}^H\mathbf{V}\tilde{\mathbf{g}}_{k_s,s}, \forall k_s. \tag{38}$$

Here, $\tilde{\mathbf{g}}_{i_j,s} = \mathbf{\Phi}_{i_j,s}\check{\mathbf{g}}_{i_j,s}$. First, we demonstrate that the computation of $\bar{\mathbf{w}}_{k_s}^{\mathrm{aslnr}}$ in (38) relies solely on real-valued matrices. According to the definition of $\bar{\mathbf{\Phi}}_{k_s,s}, \forall k_s, s$, in (33), the matrices $\bar{\mathbf{\Phi}}_{k_s,s}, \forall k_s, s$, are real-valued. According to the definition of columns of $\mathbf{V}_{\mathrm{v}}, \mathrm{v} \in \{\mathrm{x}, \mathrm{y}\}$, in (3), each column

$$\mathrm{ASLNR}_{k_s}^{\mathrm{b}} = \frac{\beta_{k_s,s}\bar{\mathbf{w}}_{k_s}^H\tilde{\mathbf{B}}_{k_s,k_s,s}\check{\mathbf{g}}_{k_s,s}\check{\mathbf{g}}_{k_s,s}^H\tilde{\mathbf{B}}_{k_s,k_s,s}^H\bar{\mathbf{w}}_{k_s}}{\bar{\mathbf{w}}_{k_s}^H\left(\sum_{j\neq s}^{S}\sum_{i=1}^{K_j}\beta_{i_j,s}\tilde{\mathbf{B}}_{k_s,i_j,s}\check{\mathbf{g}}_{i_j,s}\check{\mathbf{g}}_{i_j,s}^H\tilde{\mathbf{B}}_{k_s,i_j,s}^H + \sum_{i\neq k}^{K_s}\beta_{i_s,s}\tilde{\mathbf{B}}_{k_s,i_s,s}\check{\mathbf{g}}_{i_s,s}\check{\mathbf{g}}_{i_s,s}^H\tilde{\mathbf{B}}_{k_s,i_s,s}^H + \frac{\sigma_{k_s}^2}{p_{k_s}}\bar{\mathbf{B}}_{k_s,k_s,s}\right)\bar{\mathbf{w}}_{k_s}}. \tag{34}$$

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of $\mathbf{V}_{\mathrm{x}}$ and $\mathbf{V}_{\mathrm{y}}$ is conjugate centrosymmetric. Then, we have $\mathbf{J}_{M_{\mathrm{x}}}\mathbf{V}_{\mathrm{x}} = \mathbf{V}_{\mathrm{x}}^{*}$ and $\mathbf{J}_{M_{\mathrm{y}}}\mathbf{V}_{\mathrm{y}} = \mathbf{V}_{\mathrm{y}}^{*}$, where $\mathbf{J}_M$ is defined as

$$\mathbf{J}_M = \begin{pmatrix} 0 & \cdots & 0 & 1 \\ 0 & \cdots & 1 & 0 \\ \vdots & \ddots & \vdots & \vdots \\ 1 & \cdots & 0 & 0 \end{pmatrix}. \tag{39}$$

Then, $\mathbf{V}^H\mathbf{V}$ can be expressed as

$$\mathbf{V}^H\mathbf{V} = \mathbf{V}_{\mathrm{x}}^H\mathbf{V}_{\mathrm{x}} \otimes \mathbf{V}_{\mathrm{y}}^H\mathbf{V}_{\mathrm{y}} = \mathbf{V}_{\mathrm{x}}^T\mathbf{V}_{\mathrm{x}}^{*} \otimes \mathbf{V}_{\mathrm{y}}^T\mathbf{V}_{\mathrm{y}}^{*} = \mathbf{V}^T\mathbf{V}^{*}, \tag{40}$$

which indicates that the matrix $\mathbf{V}^H\mathbf{V}$ is real-valued. Next, we show that the diffusion vectors $\tilde{\mathbf{g}}_{i_j,s}^{\mathrm{x}}, \forall i_j, s$, are real-valued, and that the proof for $\tilde{\mathbf{g}}_{i_j,s}^{\mathrm{y}}, \forall i_j, s$, follows similarly. Recall that $\mathbf{g}_{i_j,s}^{\mathrm{x}} = \mathbf{V}_{\mathrm{x}}\tilde{\mathbf{g}}_{i_j,s}^{\mathrm{x}}$, then we have

$$[\mathbf{g}_{i_j,s}^{\mathrm{x}}]_m = \sum_{n=1}^{N_{\mathrm{x}}} [\mathbf{V}_{\mathrm{x}}]_{m,n}[\tilde{\mathbf{g}}_{i_j,s}^{\mathrm{x}}]_n, \tag{41}$$

which yields

$$[\mathbf{g}_{i_j,s}^{\mathrm{x}}]_{M_{\mathrm{x}}-m+1}^{*} = \sum_{n=1}^{N_{\mathrm{x}}} [\mathbf{V}_{\mathrm{x}}^{*}]_{M_{\mathrm{x}}-m+1,n}[\tilde{\mathbf{g}}_{i_j,s}^{\mathrm{x}}]_n^{*}. \tag{42}$$

Since $\mathbf{g}_{i_j,s}^{\mathrm{x}}$ is conjugate centrosymmetric, (41) can be rewritten as

$$[\mathbf{g}_{i_j,s}^{\mathrm{x}}]_{M_{\mathrm{x}}-m+1}^{*} = \sum_{n=1}^{N_{\mathrm{x}}} [\mathbf{V}_{\mathrm{x}}]_{m,n}[\tilde{\mathbf{g}}_{i_j,s}^{\mathrm{x}}]_n. \tag{43}$$

Because each column of $\mathbf{V}_{\mathrm{x}}$ is also conjugate centrosymmetric, we have $[\mathbf{V}_{\mathrm{x}}]_{m,n} = [\mathbf{V}_{\mathrm{x}}^{*}]_{M_{\mathrm{x}}-m+1,n}$. Combining (42) and (43) yields that

$$\mathbf{V}_{\mathrm{x}}\left(\tilde{\mathbf{g}}_{i_j,s}^{\mathrm{x}} - (\tilde{\mathbf{g}}_{i_j,s}^{\mathrm{x}})^{*}\right) = 2\jmath\mathbf{V}_{\mathrm{x}}\Im\{\tilde{\mathbf{g}}_{i_j,s}^{\mathrm{x}}\} = \mathbf{0}, \tag{44}$$

where $\Im\{\tilde{\mathbf{g}}_{i_j,s}^{\mathrm{x}}\}$ denotes the imaginary part of $\tilde{\mathbf{g}}_{i_j,s}^{\mathrm{x}}$. Denote by $\Re\{\tilde{\mathbf{g}}_{i_j,s}^{\mathrm{x}}\}$ the real part of $\tilde{\mathbf{g}}_{i_j,s}^{\mathrm{x}}$, then we have

$$\mathbf{g}_{i_j,s}^{\mathrm{x}} = \mathbf{V}_{\mathrm{x}}\left(\Re\{\tilde{\mathbf{g}}_{i_j,s}^{\mathrm{x}}\} + \jmath\Im\{\tilde{\mathbf{g}}_{i_j,s}^{\mathrm{x}}\}\right) \overset{(a)}{=} \mathbf{V}_{\mathrm{x}}\Re\{\tilde{\mathbf{g}}_{i_j,s}^{\mathrm{x}}\}, \tag{45}$$

where (a) comes from (44). This indicates that $\tilde{\mathbf{g}}_{i_j,s}, \forall i_j, s$, can be real-valued vectors. Therefore, the computation of $\bar{\mathbf{w}}_{k_s,s}^{\mathrm{aslnr}}$ in (38) relies solely on real-valued operations, and thus the computational complexity can be significantly reduced.

Next, since $\mathbf{V}^H\mathbf{V}$ in (38) does not depend on a specific UT, and it can be pre-calculated and stored. We first rewrite $\mathbf{V}^H\mathbf{V}$ as $\mathbf{V}^H\mathbf{V} = \mathbf{\Gamma}_{\mathrm{x}} \otimes \mathbf{\Gamma}_{\mathrm{y}}$, where $\mathbf{\Gamma}_{\mathrm{x}} = \mathbf{V}_{\mathrm{x}}^H\mathbf{V}_{\mathrm{x}} \in \mathbb{R}^{N_{\mathrm{x}}\times N_{\mathrm{x}}}$ and $\mathbf{\Gamma}_{\mathrm{y}} = \mathbf{V}_{\mathrm{y}}^H\mathbf{V}_{\mathrm{y}} \in \mathbb{R}^{N_{\mathrm{y}}\times N_{\mathrm{y}}}$. Then, we have

$$\begin{aligned}&\left[\mathbf{V}^H\mathbf{V}\right]_{i,j} \\ &= [\mathbf{\Gamma}_{\mathrm{x}}]_{\lfloor\frac{i-1}{N_{\mathrm{y}}}\rfloor+1,\lfloor\frac{j-1}{N_{\mathrm{y}}}\rfloor+1}\,[\mathbf{\Gamma}_{\mathrm{y}}]_{((i-1))_{N_{\mathrm{y}}}+1,((j-1))_{N_{\mathrm{y}}}+1}.\end{aligned} \tag{46}$$

Furthermore, we can obtain $\mathbf{\Gamma}_{\mathrm{x}}$ by using its first column $[\mathbf{\Gamma}_{\mathrm{x}}]_{:,1} = \mathbf{V}_{\mathrm{x}}^H[\mathbf{V}_{\mathrm{x}}]_{:,1}$, i.e.,

$$[\mathbf{\Gamma}_{\mathrm{x}}]_{m,m'} = [\mathbf{\Gamma}_{\mathrm{x}}]_{|m-m'|+1,1}. \tag{47}$$

Similarly, we can obtain $\mathbf{\Gamma}_{\mathrm{y}}$ by utilizing its first column $[\mathbf{\Gamma}_{\mathrm{y}}]_{:,1} = \mathbf{V}_{\mathrm{y}}^H[\mathbf{V}_{\mathrm{y}}]_{:,1}$. Therefore, $\mathbf{V}^H\mathbf{V}$ can be directly obtained by using the first columns of $\mathbf{\Gamma}_{\mathrm{x}}$ and $\mathbf{\Gamma}_{\mathrm{y}}$, i.e.,

$$\begin{aligned}&\left[\mathbf{V}^H\mathbf{V}\right]_{i,j} \\ &= [\mathbf{\Gamma}_{\mathrm{x}}]_{|\lfloor\frac{i-1}{N_{\mathrm{y}}}\rfloor-\lfloor\frac{j-1}{N_{\mathrm{y}}}\rfloor|+1,1}\,[\mathbf{\Gamma}_{\mathrm{y}}]_{|((i-1))_{N_{\mathrm{y}}}-((j-1))_{N_{\mathrm{y}}}|+1,1}.\end{aligned} \tag{48}$$

To obtain $\mathbf{V}^H\mathbf{V}$, we only need to calculate $[\mathbf{\Gamma}_{\mathrm{x}}]_{:,1}$ and $[\mathbf{\Gamma}_{\mathrm{y}}]_{:,1}$, whose computational and store complexities are $\mathcal{O}(N_{\mathrm{x}}\log_2 N_{\mathrm{x}} + N_{\mathrm{y}}\log_2 N_{\mathrm{y}})$ and $\mathcal{O}(N_{\mathrm{x}} + N_{\mathrm{y}})$, respectively.

Further, according to (16), the diffusion vectors $\tilde{\mathbf{g}}_{k_s,s}^{\mathrm{v}}$ can be well approximated by using $\tilde{\mathbf{g}}_{\mathrm{v}}(\xi_{k_s,s}^{\mathrm{v,min}})$, where $\xi_{k_s,s}^{\mathrm{v,min}} \in \{\xi_{i,j}^{\mathrm{v}} | i = 0, 1, \ldots, N_{\mathrm{v}} - 1, j = 0, 1, \ldots, N_{\mathrm{v}}' - 1\}$ denotes the sampled point closest to the actual directional cosine $q_{k_s,s}^{\mathrm{v}}$. The real-valued vector $\mathbf{V}^H\mathbf{V}\tilde{\mathbf{g}}_{k_s,s}$ can be expressed as

$$\left[\mathbf{V}^H\mathbf{V}\tilde{\mathbf{g}}_{k_s,s}\right]_i = \left[\mathbf{\Gamma}_{\mathrm{x}}\tilde{\mathbf{g}}_{\mathrm{x}}(\xi_{k_s,s}^{\mathrm{x,min}})\right]_m \left[\mathbf{\Gamma}_{\mathrm{y}}\tilde{\mathbf{g}}_{\mathrm{y}}(\xi_{k_s,s}^{\mathrm{y,min}})\right]_n, \tag{49}$$

where $m = \lfloor\frac{i-1}{N_{\mathrm{y}}}\rfloor + 1$, $n = ((i-1))_{N_{\mathrm{y}}} + 1$, and $\left[\mathbf{\Gamma}_{\mathrm{v}}\tilde{\mathbf{g}}_{\mathrm{v}}(\xi_{k_s,s}^{\mathrm{v,min}})\right]_m$ is given by

$$\left[\mathbf{\Gamma}_{\mathrm{v}}\tilde{\mathbf{g}}_{\mathrm{v}}(\xi_{k_s,s}^{\mathrm{v,min}})\right]_m = \sum_{m'=1}^{N_{\mathrm{v}}} [\mathbf{\Gamma}_{\mathrm{v}}]_{m,m'}\left[\tilde{\mathbf{g}}_{\mathrm{v}}(\xi_{k_s,s}^{\mathrm{v,min}})\right]_{((m'-1))_{N_{\mathrm{v}}}+1}. \tag{50}$$

Therefore, according to (47), (49), (50), and (17), $\mathbf{V}^H\mathbf{V}\tilde{\mathbf{g}}_{k_s,s}, \forall k_s, s$ can be directly obtained with a computational complexity $\mathcal{O}(N_{\mathrm{x}}'N_{\mathrm{x}}^2 + N_{\mathrm{y}}'N_{\mathrm{y}}^2 + N'N)$ and a store complexity $\mathcal{O}(N'N)$, where $N' = N_{\mathrm{x}}'N_{\mathrm{y}}'$.

Notice that $\mathbf{V}_{\mathrm{v}}^H\mathbf{V}_{\mathrm{v}}$ can be rewritten as

$$\left[\mathbf{V}_{\mathrm{v}}^H\mathbf{V}_{\mathrm{v}}\right]_{i_{\mathrm{v}},j_{\mathrm{v}}} = \Upsilon_{\mathrm{v}}\left(\frac{i_{\mathrm{v}} - j_{\mathrm{v}}}{N_{\mathrm{v}}}\right), \tag{51}$$

where $\Upsilon_{\mathrm{v}}(x) = \frac{\sin(\pi M_{\mathrm{v}}x)}{M_{\mathrm{v}}\sin(\pi x)}$. As $|i_{\mathrm{v}} - j_{\mathrm{v}}|$ increases, $|[\mathbf{V}_{\mathrm{v}}^H\mathbf{V}_{\mathrm{v}}]_{i_{\mathrm{v}},j_{\mathrm{v}}}|$ decreases. Let $\bar{\mathbf{g}}_{k_s,m,s} = \bar{\mathbf{\Phi}}_{k_s,s}^H\mathbf{V}^H\mathbf{V}\tilde{\mathbf{g}}_{m,s} = \bar{\mathbf{\Phi}}_{k_s,s}^H\mathbf{V}^H\mathbf{V}\mathbf{\Phi}_{m,s}\check{\mathbf{g}}_{m,s}$, we can rewrite (38) as

$$\begin{aligned}\bar{\mathbf{w}}_{k_s}^{\mathrm{aslnr}} = \Bigg(&\beta_{k_s,s}\bar{\mathbf{g}}_{k_s,k_s,s}\bar{\mathbf{g}}_{k_s,k_s,s}^H + \frac{\sigma_{k_s}^2}{p_{k_s}}\bar{\mathbf{B}}_{k_s,k_s,s} \\ &+ \sum_{m\neq k_s}^{K}\beta_{m,s}\bar{\mathbf{g}}_{k_s,m,s}\bar{\mathbf{g}}_{k_s,m,s}^H\Bigg)^{-1}\bar{\mathbf{g}}_{k_s,k_s,s}, \forall k_s,\end{aligned} \tag{52}$$

Then, we define $\mathcal{U}_{k_s} = \{m | ||\bar{\mathbf{g}}_{k_s,m,s}||_\infty > \varepsilon\}$, where ε denotes the threshold to exclude the UTs whose interference is negligible. We note that the threshold ε serves to balance the trade-off between computational complexity and the performance achieved by the precoder. When ε is sufficiently small, the matrix $\sum_{m=1,m\notin\mathcal{U}_{k_s},m\neq k_s}^{K}\beta_{m,s}\bar{\mathbf{g}}_{k_s,m,s}\bar{\mathbf{g}}_{k_s,m,s}^H$ can be approximated as

$$\sum_{m=1,m\notin\mathcal{U}_{k_s},m\neq k_s}^{K}\beta_{m,s}\bar{\mathbf{g}}_{k_s,m,s}\bar{\mathbf{g}}_{k_s,m,s}^H \approx \mathbf{0}. \tag{53}$$

Then, the interference terms in the beam domain precoder design can be approximated as

$$\sum_{m=1,m\neq k_s}^{K}\beta_{m,s}\bar{\mathbf{g}}_{k_s,m,s}\bar{\mathbf{g}}_{k_s,m,s}^H \approx \sum_{m\in\mathcal{U}_{k_s}\setminus\{k_s\}}\beta_{m,s}\bar{\mathbf{g}}_{k_s,m,s}\bar{\mathbf{g}}_{k_s,m,s}^H. \tag{54}$$

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Therefore, (52) is approximated as

$$\bar{\mathbf{w}}_{k_s}^{\text{aslnr}} \approx \left(\beta_{k_s,s} \bar{\mathbf{g}}_{k_s,k_s,s} \bar{\mathbf{g}}_{k_s,k_s,s}^H + \frac{\sigma_{k_s}^2}{p_{k_s}} \bar{\mathbf{B}}_{k_s,k_s,s} + \sum_{m \in \mathcal{U}_{k_s} \setminus \{k_s\}} \beta_{m,s} \bar{\mathbf{g}}_{k_s,m,s} \bar{\mathbf{g}}_{k_s,m,s}^H \right)^{-1} \bar{\mathbf{g}}_{k_s,k_s,s}, \forall k_s. \quad (55)$$

For brevity, we denote the average of $\bar{N}_{k_s,s}^{\text{dia}}$ as $\bar{N}_{\text{dl}} = \frac{1}{K} \sum_{s=1}^{S} \sum_{k=1}^{K_s} \bar{N}_{k_s,s}^{\text{dia}}$. The design complexity of the beam domain DL precoder $\{\bar{\mathbf{w}}_{k_s}^{\text{aslnr}}\}_{k \in \mathcal{K}_s, s \in \mathcal{S}}$ is given by $\mathcal{O}\left(\sum_{s=1}^{S} \sum_{k=1}^{K_s} \left(\frac{|\mathcal{U}_{k_s}|}{2} \bar{N}_{\text{dl}}^2 + \bar{N}_{\text{dl}}^3 + \bar{N}_{\text{dl}}^2 \right)\right)$.

For comparison, we provide the design complexity of the space domain DL precoder. The design of the space domain precoder is presented in (25), when the number of UTs in the SATCOM system is less than the number of satellite antennas, the matrix inversion in the dimension of the number of satellite antennas can be transformed into a matrix inversion in the dimension of the number of UTs with lower complexity. Therefore, (25) can be rewritten as

$$\begin{aligned} \mathbf{w}_{k_s}^{\text{aslnr}} &= \frac{1}{\eta_{k_s}^{\text{aslnr}}} \left[\left(\mathbf{G}_s \mathbf{G}_s^H + \frac{\sigma_{k_s}^2}{p_{k_s}} \mathbf{I} \right)^{-1} \mathbf{G}_s \mathbf{A}_s^{-\frac{1}{2}} \right]_{:,k_s} \\ &= \frac{1}{\eta_{k_s}^{\text{aslnr}}} \left[\mathbf{G}_s \left(\mathbf{G}_s^H \mathbf{G}_s + \frac{\sigma_{k_s}^2}{p_{k_s}} \mathbf{I} \right)^{-1} \mathbf{A}_s^{-\frac{1}{2}} \right]_{:,k_s}, \forall k_s, \end{aligned} \quad (56)$$

where $\mathbf{G}_s = [\mathbf{g}_{1,s}, \mathbf{g}_{2,s}, \ldots, \mathbf{g}_{K,s}] \times \mathbf{A}_s^{\frac{1}{2}} \in \mathbb{C}^{M \times K}$, and $\mathbf{A}_s = \text{diag}(\beta_{1,s}, \beta_{2,s}, \ldots, \beta_{K,s}) \in \mathbb{R}^{K \times K}$. The design complexity of the space domain DL precoders $\{\mathbf{w}_{k_s}^{\text{aslnr}}\}_{k \in \mathcal{K}_s, s \in \mathcal{S}}$ is given by $\mathcal{O}\left(\sum_{s=1}^{S} K_s K^3 + \frac{3}{2} M K^2\right)$.

B. Low-Complexity Implementation

The implementation step refers to applying the designed precoders to the transmit data streams to generate the transmit signals at each satellite. By using the designed beam structured precoder, $\mathbf{x}_s$ in (18) can be rewritten as

$$\mathbf{x}_s = (\mathbf{V}_{\text{x}} \otimes \mathbf{V}_{\text{y}}) \mathbf{s}_s = \text{vec}\left(\mathbf{V}_{\text{y}} \mathbf{S}_s \mathbf{V}_{\text{x}}^T\right), \quad (57)$$

where $\mathbf{s}_s = \sum_{k=1}^{K_s} \bar{\mathbf{\Phi}}_{k_s,s} \sqrt{p_{k_s}} \bar{\mathbf{w}}_{k_s} s_{k_s}$, and $\mathbf{S}_s = \text{uvec}(\mathbf{s}_s) \in \mathbb{C}^{N_{\text{y}} \times N_{\text{x}}}$. Therefore, the complexity for generating the transmit signals is calculated as $\mathcal{O}\left(\sum_{s=1}^{S} N \log_2 N_{\text{y}} + M_{\text{y}} N_{\text{x}} \log_2 N_{\text{x}} + K_s \bar{N}_{\text{dl}}\right)$. The complexity for generating the transmit signal by using the space domain precoder is given by $\mathcal{O}\left(\sum_{s=1}^{S} K_s M\right)$.

For the computation of the total complexity, we assume that the DL precoders are designed once within 100 milliseconds for 4 resource blocks (14 OFDM symbols per milliseconds and 12 subcarriers per recourse block) [52], and implemented to generate the signal for UTs. Therefore, the total complexity of the proposed sCSI based beam structured DL precoder within 100 milliseconds is given by $\mathcal{O}\Big(\Big(\sum_{s=1}^{S} \sum_{k=1}^{K_s} \frac{|\mathcal{U}_{k_s}|}{2} \bar{N}_{\text{dl}}^2 + \bar{N}_{\text{dl}}^3 + \bar{N}_{\text{dl}}^2\Big) + \Big(\sum_{s=1}^{S} N \log_2 N_{\text{y}} + M_{\text{y}} N_{\text{x}} \log_2 N_{\text{x}} + K_s \bar{N}_{\text{dl}}\Big) \times 100 \times 4 \times 14 \times 12\Big)$. The total complexity of the sCSI based space domain DL precoder within 100 ms is given by $\mathcal{O}\Big(\Big(\sum_{s=1}^{S} K_s K^3 + \frac{3}{2} M K^2\Big) + 100 \times 4 \times 14 \times 12 \times \sum_{s=1}^{S} K_s M\Big)$.

TABLE II
SIMULATION PARAMETERS

Parameter	Value
Number of UTs K	256
Number of satellites S	4
Orbit altitude H	600 km
Earth Radius R_{e}	6378 km
Bandwidth B	10 MHz
Maximum nadir angle $\vartheta_{\max}$	45°
Number of satellite antennas $M_{\text{x}} \times M_{\text{y}}$	20×20
Antenna spacing d_{x} (d_{y})	$\frac{\lambda}{2}$
Carrier frequency f_c	2 GHz
Noise Temperature T_n	273 K
Transmit Power of satellite s $P_s = P, \forall s$	5 dBW $\sim$ 25 dBW

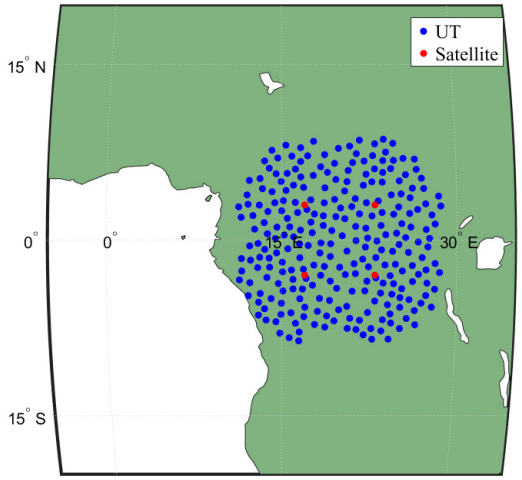


Fig. 3. UT distribution.

V. SIMULATION RESULTS

In this section, simulation results are provided to verify the performance of the proposed beam structured DL precoders for massive MIMO multiple LEO SATCOM. Simulation parameters are listed in Table II. We adopt a LEO altitude of $H = 600$ km as studied in 3rd Generation Partnership Project (3GPP) NTN [53], and four satellites operating at lower frequency bands, e.g., L/S bands, at the latitude/longitude pairs $(3°, 17°), (3°, 23°), (-3°, 17°), (-3°, 23°)$. UTs are uniformly distributed within the satellites' coverage area, as presented in Fig. 3. The inter-satellite distance is around 700 km, which falls within a reasonable range [54]. Denote by $R_{\text{sat}} = H + R_{\text{e}}$ the satellite orbit radius, where H is the orbit altitude of each satellite, and R_{e} is the earth radius. The elevation angle can be calculated as $\alpha_{k_s,s} = \cos^{-1}\left(\frac{R_{\text{sat}} \sin(\vartheta_{k_s,s})}{R_{\text{e}}}\right)$ [55]. Denote by $D_{k_s,s}$ the distance between UT k_s and satellite s, which is $D_{k_s,s} = -\sin(\alpha_{k_s,s}) R_{\text{e}} + \sqrt{2 R_{\text{e}} H + (\sin(\alpha_{k_s,s}) R_{\text{e}})^2 + H^2}$ [53]. The

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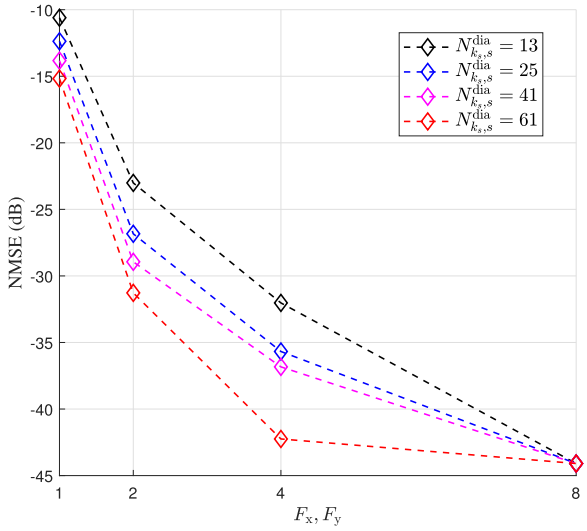


Fig. 4. NMSE performance of the DL BBCM versus the fine factor for different numbers of beams.

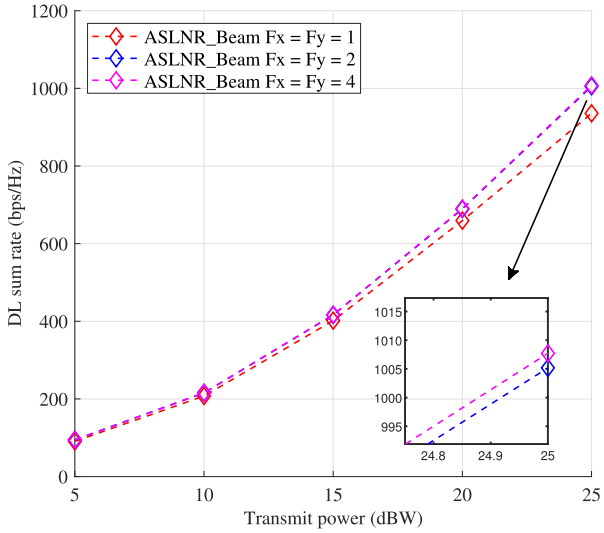


Fig. 5. DL ergodic sum rate performance of the proposed beam structured precoder design for different fine factors.

antenna gain of satellite s for UT k_s is denoted as $G_{\mathrm{sat}}(q_{k_s,s}^{\mathrm{x}}, q_{k_s,s}^{\mathrm{y}})$ by considering the radiation power pattern [56]. We denote G_{ut} as the antenna gain at the UT side. The complex multipath channel gains are simulated by using the exponential power delay profile [24], [53]. The system bandwidth is set as $B = 10$ MHz [57]. The noise temperature is $T_n = 273$ K [23].The ionospheric scintillation loss is set as 2 dB approximately [53]. In addition, the shadow fading and path loss for different elevation angles are adopted based on the dense urban scenario in [53].

We first evaluate the normalized mean-squared error (NMSE) performance of our proposed DL BBCM, as depicted in Fig. 4, with different fine factors and numbers of beams. The NMSE is calculated as

$$\mathrm{NMSE} = \frac{\sum_{s=1}^{S}\sum_{j=1}^{S}\sum_{k=1}^{K_j}\left\|\mathbf{g}_{k_j,s} - \mathbf{V}_{k_j,s}\mathbf{\Phi}_{k_j,s}\check{\mathbf{g}}_{k_j,s}\right\|^2}{\sum_{s=1}^{S}\sum_{j=1}^{S}\sum_{k=1}^{K_j}\left\|\mathbf{g}_{k_j,s}\right\|^2}. \quad (58)$$

From the figure, we can observe that a smaller beam set with a higher fine factor yields a better NMSE performance compared with a larger beam set with a smaller fine factor, confirming the sparsity of the beam domain satellite channel. To balance the complexity and modeling accuracy, the subsequent simulations adopt $F_{\mathrm{x}} = F_{\mathrm{y}} = 2$ and $N_{k_s,s}^{\mathrm{dia}} = 25$ when establishing the BBCM.

We evaluate the system performance by using the DL ergodic sum rate, which reflects the effect of the asynchronization, and is given by [43]

$$\mathcal{R}_{\mathrm{dl}}^{\mathrm{sum}} = \sum_{s=1}^{S}\sum_{k=1}^{K_s}\mathbb{E}\left\{\log\left(1 + \frac{p_{k_s}\left|\mathbf{w}_{k_s}^H\mathbf{h}_{k_s,s}\right|^2}{\sum_{i\neq k}^{K_s} p_{i_s}\left|\mathbf{w}_{i_s}^H\mathbf{h}_{k_s,s}\right|^2 + R_{k_s}}\right)\right\}, \quad (59)$$

where $R_{k_s} = \sum_{j\neq s}^{S}\sum_{i=1}^{K_j}\beta_{k_s,j}|\mathbf{w}_{ij}^H\mathbf{g}_{k_s,j}|^2 p_{i_j} + \sigma_{k_s}^2$.

Fig. 5 presents the DL sum rate performance of the proposed beam structured precoder design for different fine factors. The

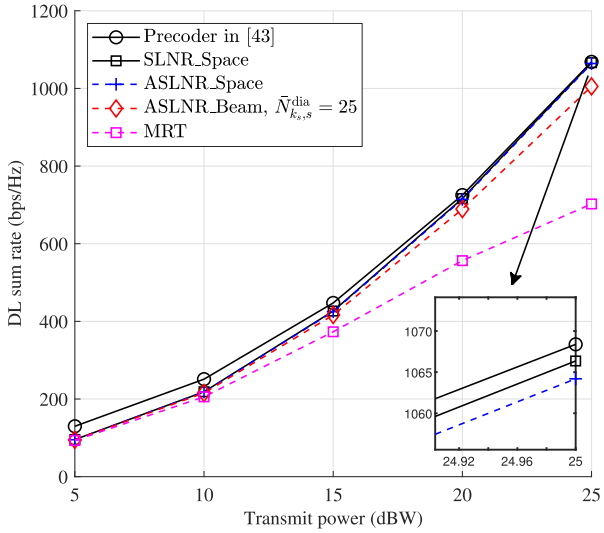


Fig. 6. DL ergodic sum rate performance of the proposed space domain precoder design and the beam structured precoder design compared to the precoding strategy in [43] and the maximum-ratio transmission (MRT) scheme.

DL ergodic sum rate is calculated by replacing $\mathbf{w}_{k_s}$ in (59) with $\bar{\mathbf{V}}_{k_s,s}\bar{\mathbf{w}}_{k_s}^{\mathrm{aslnr}}$. It can be observed that the beam structured precoder design with $F_{\mathrm{x}} = F_{\mathrm{y}} = 1$ achieve the worst performance, while configurations with $F_{\mathrm{x}} = F_{\mathrm{y}} = 2$ and $F_{\mathrm{x}} = F_{\mathrm{y}} = 4$ show a significant performance improvement. The sum rate difference between these two configurations is negligible. In the following simulations, the fine factors related to the x-axis and y-axis are set as 2.

Fig. 6 shows the DL ergodic sum rate performance of the proposed space domain precoder design and the beam structured precoder design compared to the precoding strategy in [43] and the MRT scheme. It can be observed that the performance difference between iCSI and sCSI based space domain

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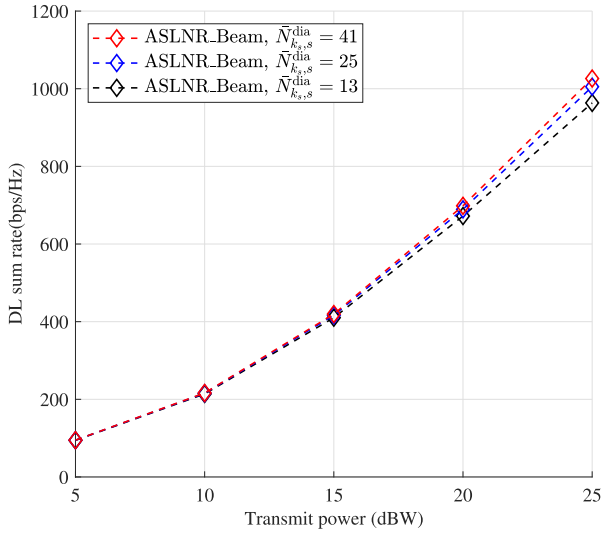


Fig. 7. DL ergodic sum rate performance of the proposed beam structured precoder design for different numbers of beams.

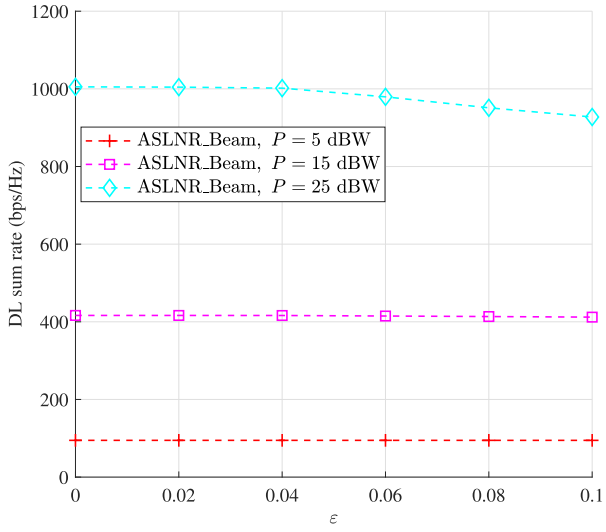


Fig. 8. DL ergodic sum rate performance of the proposed beam structured precoder design ($\bar{N}^{\text{dia}}_{k_s,s} = 25$) for different values of threshold ϵ.

precoders is negligible, and they can achieve a performance comparable to that of the precoding strategy proposed in [43] based on the ergodic sum rate upper bound maximization. The proposed beam structured precoder can achieve over 95% of the performance of the space domain precoder. Moreover, the MRT scheme, i.e., $\mathbf{w}_{k_s,s} = \sqrt{p_{k,s}}\mathbf{g}^*_{k_s,s}$, corresponds to equation (36) in [21] and exhibits the poorest performance.

Fig. 7 shows the DL sum rate performance of the proposed beam structured precoder design for different numbers of selected beams. The sum rate performance for different values of $\bar{N}^{\text{dia}}_{k_s,s}$ is achieved by the beam structured precoder with beam index set $\bar{\mathcal{N}}^{\text{dia}}_{k_s,s}$. In the simulation, the values of $\bar{N}^{\text{dia}}_{k_s,s}$ are set as 13, 25 and 41, respectively. We can observe that the performance can be improved by selecting more beams according to Theorem 2 when the beam domain channels of different UTs are overlapping.

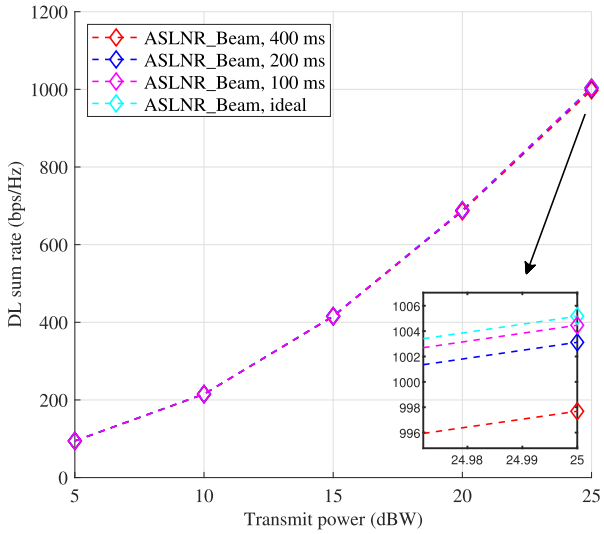


Fig. 9. DL ergodic sum rate performance of the proposed beam structured precoder design ($\bar{N}^{\text{dia}}_{k_s,s} = 25$) for different update periods.

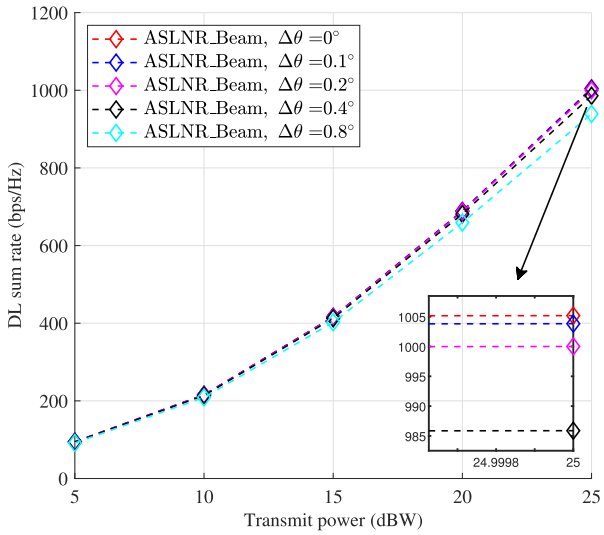


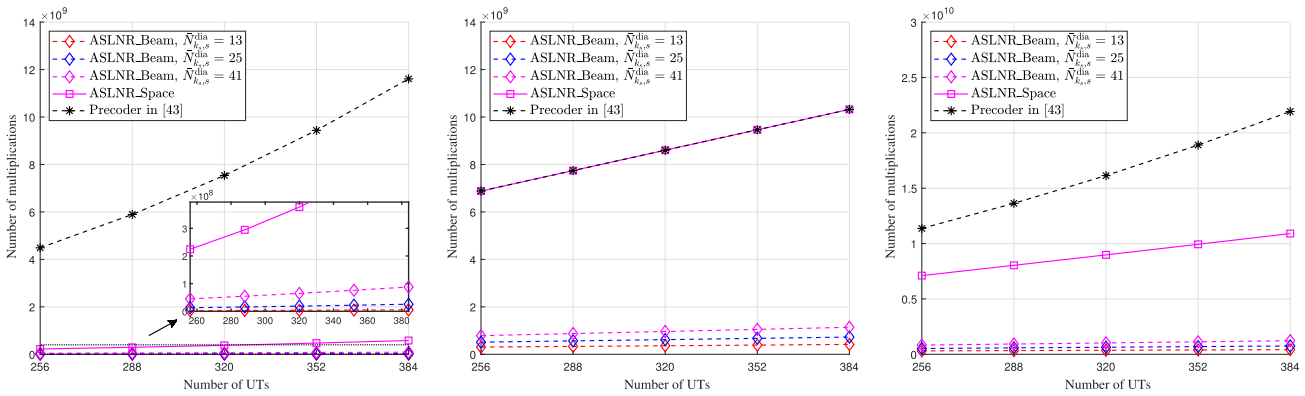
Fig. 10. DL ergodic sum rate performance of the proposed beam structured precoder design ($\bar{N}^{\text{dia}}_{k_s,s} = 25$) for different inclination angles.

The DL sum rate performance of the proposed beam structured precoder design with $\bar{N}^{\text{dia}}_{k_s,s} = 25$ for different values of the threshold ϵ is depicted in Fig. 8. It can be observed that the DL sum rate performance degrades with the threshold ε increasing, especially in the high transmit power case. By choosing an appropriate threshold to exclude the UTs whose interference is negligible, the design complexity can be reduced without significantly sacrificing performance.

Fig. 9 shows the DL sum rate performance of the proposed beam structured precoder design under different update periods. We can observe that the sum rate performance degrade of the beam structured precoder design is negligible with the update period of 200 ms.

Fig. 10 presents the DL sum rate performance of the proposed beam structured precoder design for different values

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(a) Design complexity (b) Implementation complexity (c) Total complexity

Fig. 11. Complexities of the proposed space domain and beam structured precoders compared to the precoding strategy in [43] under different numbers of selected beams for different UT numbers.

of $\Delta\theta$, which refers to the array inclination angles relative to the x- and y- axes. We can observe that the proposed beam structured precoder performs effectively for $\Delta\theta \leq 0.2°$, which corresponds to UT location errors on the order of a few kilometers.

Fig. 11 presents the design complexity, implementation complexity, and total complexity of our proposed designs compared to the precoder in [43] for different UT numbers with different beam sets. Specifically, Fig. 11a shows that the design complexity of our proposed precoder designs is much lower than that of the precoder in [43]. For the beam structured precoder, the design complexity increases as $\bar{N}^{\text{dia}}_{k_s,s}$ grows, while it remains much lower than that of the space domain counterpart. Fig. 11b shows that the implementation complexity of the space domain precoder is identical to that of the precoder in [43], as they share the same vector size. Moreover, the beam structured precoder design has a much lower implementation complexity. Further, Fig. 11c shows that the precoder in [43] and the proposed space domain precoder exhibit a linear increase in the total complexity as the number of UTs grows. In contrast, the beam structured precoder under three configurations ($\bar{N}^{\text{dia}}_{k_s,s}=13$, $\bar{N}^{\text{dia}}_{k_s,s}=25$, and $\bar{N}^{\text{dia}}_{k_s,s}=41$) shows consistently low complexity. When the number of UTs is large, using beam structured precoders can avoid significant complexity compared to space domain precoders, without significant performance degradation.

VI. Conclusion

In this paper, we have investigated the DL precoder design for massive MIMO multi-satellite communications. We first established a DL BBCM by using sampled array response vectors. Then, we presented the closed-form sCSI based DL precoder for the massive MIMO multi-satellite system to maximize the ASLNR. With the established BBCM, we showed that the design of space domain precoder can be transformed into that of a lower-dimensional beam domain vector, and the resulting space domain precoder is beam structured. Furthermore, we proposed a low-complexity design and implementation for the proposed beam structured precoder. Simulation results verified the effectiveness of our proposed DL precoder design for massive MIMO multiple LEO SATCOM. Future work can extend the beam structured precoder design to an ultra massive MIMO transmission for 6G terrestrial cellular systems with higher carrier frequencies and sparse channel conditions.

Appendix A
Proof of Theorem 1

We first define $\mathcal{N}^{\text{dia}}_s = \mathcal{N}^{\text{dia}}_{1,s} \cup \mathcal{N}^{\text{dia}}_{2,s} \cup \cdots \cup \mathcal{N}^{\text{dia}}_{K_s,s}$ with $|\mathcal{N}^{\text{dia}}_s| = N^{\text{dia}}_s$ and denote $\breve{\mathbf{V}}_s = [\mathbf{V}]_{:,\mathcal{N}^{\text{dia}}_s} \in \mathbb{C}^{M\times N^{\text{dia}}_s}$, then (9) can be rewritten as

$$\mathbf{g}_{k_s,s} = \breve{\mathbf{V}}_s \breve{\mathbf{g}}_{k_s,s}, \tag{60}$$

where $\breve{\mathbf{g}}_{k_s,s} = [\tilde{\mathbf{g}}_{k_s,s}]_{\mathcal{N}^{\text{dia}}_s}$. We define $\breve{\mathbf{w}}_{k_s} = \breve{\mathbf{V}}^H_s \mathbf{w}_{k_s}$, and substitute $\mathbf{g}_{k_s,s}$ in (60) into (24), then we can rewrite (24) as

$$\text{ASLNR}_{k_s}(\breve{\mathbf{w}}_{k_s}) = \frac{\beta_{k_s,s}\left|\breve{\mathbf{w}}^H_{k_s}\breve{\mathbf{g}}_{k_s,s}\right|^2}{\sum_{i\neq k}^{K_s}\beta_{i_s,s}\left|\breve{\mathbf{w}}^H_{k_s}\breve{\mathbf{g}}_{i_s,s}\right|^2 + \breve{L}_{k_s} + \sigma^2_{k_s}/p_{k_s}}, \tag{61}$$

where $\breve{L}_{k_s} = \sum_{j\neq s}^{S}\sum_{i=1}^{K_j}\beta_{ij,s}|\breve{\mathbf{g}}^H_{ij,s}\breve{\mathbf{w}}_{k_s}|^2$. For the optimal $\breve{\mathbf{w}}^{\text{opt}}_{k_s}$, there exists the corresponding $\mathbf{w}^{\text{opt}}_{k_s}$ such that $\breve{\mathbf{w}}^{\text{opt}}_{k_s} = \breve{\mathbf{V}}^H_s \mathbf{w}^{\text{opt}}_{k_s}$ [58, Lemma 1]. Define $\mathbf{n}_{k_s,s} \in \mathbb{R}^{N^{\text{dia}}_s\times 1}$ as

$$[\mathbf{n}_{k_s,s}]_i = \begin{cases} 1, & [\mathcal{N}^{\text{dia}}_s]_i \in \mathcal{N}^{\text{dia}}_{k_s,s}, \\ 0, & \text{otherwise.} \end{cases} \tag{62}$$

Define $\breve{\mathbf{w}}^{\text{opt}}_{k_s} = \breve{\mathbf{w}}^{\text{opt}}_{k_s} \odot \mathbf{n}_{k_s,s}$. As $\breve{\mathbf{w}}^{\text{opt}}_{k_s}$ is the optimal solution, then we can obtain that

$$\text{ASLNR}_{k_s}\left(\breve{\mathbf{w}}^{\text{opt}}_{k_s}\right) \leq \text{ASLNR}_{k_s}\left(\breve{\mathbf{w}}^{\text{opt}}_{k_s}\right). \tag{63}$$

On the other hand, because $\mathcal{N}_{k,s} \cap \mathcal{N}_{i,s} = \emptyset, \forall s,k,i$, when calculating ASLNR with (61) by replacing $\breve{\mathbf{w}}^{\text{opt}}_{k_s}$ with $\breve{\mathbf{w}}^{\text{opt}}_{k_s}$, the numerator remains unchanged, while the denominator may either decrease or stay the same. Then, we have

$$\text{ASLNR}_{k_s}\left(\breve{\mathbf{w}}^{\text{opt}}_{k_s}\right) \geq \text{ASLNR}_{k_s}\left(\breve{\mathbf{w}}^{\text{opt}}_{k_s}\right). \tag{64}$$

Therefore, we can obtain

$$\text{ASLNR}_{k_s}\left(\breve{\mathbf{w}}^{\text{opt}}_{k_s}\right) = \text{ASLNR}_{k_s}\left(\breve{\mathbf{w}}^{\text{opt}}_{k_s}\right), \tag{65}$$

which indicates that we can focus on the design of elements in $[\breve{\mathbf{w}}^{\text{opt}}_{k_s}]_{\mathcal{N}^{\text{dia}}_{k_s,s}}$. Therefore, we have that $\breve{\mathbf{w}}_{k_s} = \breve{\mathbf{V}}^H_{k_s,s}\mathbf{w}_{k_s}, \forall k_s$.

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For $\check{\mathbf{w}}_{k_s}^{\mathrm{opt}} = \check{\mathbf{V}}_{k_s,s}^H \mathbf{w}_{k_s}^{\mathrm{opt}}$, a minimum-norm solution of $\mathbf{w}_{k_s}^{\mathrm{opt}}$ is expressed as [59, Corollary 1]

$$\begin{aligned}\mathbf{w}_{k_s}^{\mathrm{opt}} &= \check{\mathbf{V}}_{k_s,s}\left(\check{\mathbf{V}}_{k_s,s}^H \check{\mathbf{V}}_{k_s,s}\right)^{-1} \check{\mathbf{w}}_{k_s}^{\mathrm{opt}} \\ &\overset{(a)}{=} \check{\mathbf{V}}_{k_s,s}\check{\mathbf{w}}_{k_s}^{\mathrm{opt}},\end{aligned} \tag{66}$$

where (a) comes from $\lim_{M\to\infty} \check{\mathbf{V}}_{k_s,s}^H \check{\mathbf{V}}_{k_s,s} = \mathbf{I}$. In addition, since $\|\mathbf{w}_{k_s}^{\mathrm{opt}}\|^2 = 1$, we have $\|\check{\mathbf{w}}_{k_s}^{\mathrm{opt}}\|^2 = 1$. The optimal solution of $\check{\mathbf{w}}_{k_s}$ is given by

$$\begin{aligned}&\check{\mathbf{w}}_{k_s}^{\mathrm{opt}} \\ &= \arg\max_{\check{\mathbf{w}}_{k_s}} \frac{\beta_{k_s,s}\left|\check{\mathbf{w}}_{k_s}^H \check{\mathbf{g}}_{k_s,s}\right|^2}{\sum_{i\neq k}^{K_s}\beta_{i_s,s}\left|\check{\mathbf{w}}_{k_s}^H \check{\mathbf{V}}_{k_s,s}^H \check{\mathbf{V}}_{i_s,s}\check{\mathbf{g}}_{i_s,s}\right|^2 + \check{L}_{k_s} + \frac{\sigma_{k_s}^2}{p_{k_s}}},\end{aligned} \tag{67}$$

where $\check{L}_{k_s} = \sum_{j\neq s}^{S}\sum_{i=1}^{K_j}\beta_{i_j,s}|\check{\mathbf{g}}_{i_j,s}^H \check{\mathbf{V}}_{i_j,s}^H \check{\mathbf{V}}_{k_s,s}\check{\mathbf{w}}_{k_s}|^2$. This completes the proof.

Appendix B
Proof of Theorem 2

Since the precoding design under the smaller beam index set $\mathcal{N}_{k_s,s}^{\mathrm{dia}}$ is a special case of that the larger beam index set $\bar{\mathcal{N}}_{k_s,s}^{\mathrm{dia}}$, we can directly obtain that

$$\mathrm{ASLNR}_{k_s}^{\mathrm{b}}\left(\check{\mathbf{w}}_{k_s}^{\mathrm{aslnr}}, \bar{\mathcal{N}}_{k_s,s}^{\mathrm{dia}}\right) \geq \mathrm{ASLNR}_{k_s}^{\mathrm{b}}\left(\check{\mathbf{w}}_{k_s}^{\mathrm{aslnr}}, \mathcal{N}_{k_s,s}^{\mathrm{dia}}\right). \tag{68}$$

Next, we derive the condition under which the equality in (68) holds. We define the optimal solution as $\check{\mathbf{w}}_{k_s}^{\diamondsuit}$ with the beam index set $\bar{\mathcal{N}}_{k_s,s}^{\mathrm{dia}}$, and define the $\bar{N}_{k_s,s}^{\mathrm{dia}} \times \bar{N}_{k_s,s}^{\mathrm{dia}}$ real-valued diagonal matrices $\mathbf{P}_{k_s}^{\circ}$ and $\mathbf{P}_{k_s}^{\Delta}$ as

$$\left[\mathbf{P}_{k_s}^{\circ}\right]_{j,j} = \begin{cases} 1, & [\bar{\mathcal{N}}_{k_s,s}^{\mathrm{dia}}]_j \in \mathcal{N}_{k_s,s}^{\mathrm{dia}}, \\ 0, & \text{otherwise}, \end{cases} \tag{69}$$

and

$$\left[\mathbf{P}_{k_s}^{\Delta}\right]_{j,j} = \begin{cases} 1, & [\bar{\mathcal{N}}_{k_s,s}^{\mathrm{dia}}]_j \in \bar{\mathcal{N}}_{k_s,s}^{\mathrm{dia}} \setminus \mathcal{N}_{k_s,s}^{\mathrm{dia}}, \\ 0, & \text{otherwise}, \end{cases} \tag{70}$$

respectively, which satisfy $\mathbf{P}_{k_s}^{\circ} + \mathbf{P}_{k_s}^{\Delta} = \mathbf{I}$, $\mathbf{P}_{k_s}^{\circ}\mathbf{P}_{k_s}^{\Delta} = \mathbf{0}$, and $\mathbf{P}_{k_s}^{\Delta}\mathbf{P}_{k_s}^{\circ} = \mathbf{0}$. Then, we have

$$\check{\mathbf{w}}_{k_s}^{\diamondsuit} = \check{\mathbf{w}}_{k_s}^{\circ} + \check{\mathbf{w}}_{k_s}^{\Delta}, \tag{71}$$

where $\check{\mathbf{w}}_{k_s}^{\circ} = \mathbf{P}_{k_s}^{\circ}\check{\mathbf{w}}_{k_s}^{\diamondsuit}$, and $\check{\mathbf{w}}_{k_s}^{\Delta} = \mathbf{P}_{k_s}^{\Delta}\check{\mathbf{w}}_{k_s}^{\diamondsuit}$. It can be verified that $||\check{\mathbf{w}}_{k_s}^{\circ}||^2 \leq ||\check{\mathbf{w}}_{k_s}^{\diamondsuit}||^2 = 1$, and $||\check{\mathbf{w}}_{k_s}^{\circ}||^2$ can be scaled such that $||\check{\mathbf{w}}_{k_s}^{\circ}||^2 = 1$ without affecting the value of ASLNR. Therefore, $\check{\mathbf{w}}_{k_s}^{\circ}$ can satisfy the power constraint. For the numerator of (29), we can obtain that

$$\begin{aligned}&(\check{\mathbf{w}}_{k_s}^{\diamondsuit})^H \mathbf{Q}_{k_s,s}\left[\tilde{\mathbf{g}}_{k_s,s}\right]_{\mathcal{N}_{k_s,s}^{\mathrm{dia}}}\left((\check{\mathbf{w}}_{k_s}^{\diamondsuit})^H \mathbf{Q}_{k_s,s}\left[\tilde{\mathbf{g}}_{k_s,s}\right]_{\mathcal{N}_{k_s,s}^{\mathrm{dia}}}\right)^H \\ &= (\check{\mathbf{w}}_{k_s}^{\circ})^H \mathbf{Q}_{k_s,s}[\tilde{\mathbf{g}}_{k_s,s}]_{\mathcal{N}_{k_s,s}^{\mathrm{dia}}}\left((\check{\mathbf{w}}_{k_s}^{\circ})^H \mathbf{Q}_{k_s,s}[\tilde{\mathbf{g}}_{k_s,s}]_{\mathcal{N}_{k_s,s}^{\mathrm{dia}}}\right)^H,\end{aligned} \tag{72}$$

where $\mathbf{Q}_{k_s,s} = [\mathbf{I}_N]_{\bar{\mathcal{N}}_{k_s,s}^{\mathrm{dia}}, \mathcal{N}_{k_s,s}^{\mathrm{dia}}}$. The denominator of (29) is calculated as

$$\begin{aligned}(\check{\mathbf{w}}_{k_s}^{\diamondsuit})^H \tilde{\mathbf{L}}_{k_s}\check{\mathbf{w}}_{k_s}^{\diamondsuit} + \frac{\sigma_{k_s}^2}{p_{k_s}} &= (\check{\mathbf{w}}_{k_s}^{\circ})^H \tilde{\mathbf{L}}_{k_s}\check{\mathbf{w}}_{k_s}^{\circ} + (\check{\mathbf{w}}_{k_s}^{\Delta})^H \tilde{\mathbf{L}}_{k_s}\check{\mathbf{w}}_{k_s}^{\Delta} \\ &+ (\check{\mathbf{w}}_{k_s}^{\circ})^H \tilde{\mathbf{L}}_{k_s}\check{\mathbf{w}}_{k_s}^{\Delta} + (\check{\mathbf{w}}_{k_s}^{\Delta})^H \tilde{\mathbf{L}}_{k_s}\check{\mathbf{w}}_{k_s}^{\circ} + \frac{\sigma_{k_s}^2}{p_{k_s}},\end{aligned} \tag{73}$$

where $\tilde{\mathbf{L}}_{k_s}$ is given by

$$\tilde{\mathbf{L}}_{k_s} = \sum_{j\neq s}^{S}\sum_{i=1}^{K_j}\beta_{i_j,s}\mathbf{A}_{k_s,i_j,s} + \sum_{i\neq k}^{K_s}\beta_{i_s,s}\mathbf{A}_{k_s,i_s,s}, \tag{74}$$

with

$$\mathbf{A}_{k_s,i_j,s} = \tilde{\mathbf{Q}}_{k_s,i_j,s}\left[\tilde{\mathbf{g}}_{i_j,s}\right]_{\mathcal{N}_{i_j,s}^{\mathrm{dia}}}\left[\tilde{\mathbf{g}}_{i_j,s}\right]_{\mathcal{N}_{i_j,s}^{\mathrm{dia}}}^H \tilde{\mathbf{Q}}_{k_s,i_j,s}^T, \tag{75}$$

and $\tilde{\mathbf{Q}}_{k_s,i_j,s} = [\mathbf{I}_N]_{\bar{\mathcal{N}}_{k_s,s}^{\mathrm{dia}}, \mathcal{N}_{i_j,s}^{\mathrm{dia}}}$. Then, $(\check{\mathbf{w}}_{k_s}^{\circ})^H \tilde{\mathbf{L}}_{k_s}\check{\mathbf{w}}_{k_s}^{\Delta}$ in (73) can be written as

$$\begin{aligned}(\check{\mathbf{w}}_{k_s}^{\circ})^H \tilde{\mathbf{L}}_{k_s}\check{\mathbf{w}}_{k_s}^{\Delta} &= \sum_{j\neq s}^{S}\sum_{i=1}^{K_j}\beta_{i_j,s}(\check{\mathbf{w}}_{k_s}^{\circ})^H \mathbf{A}_{k_s,i_j,s}\check{\mathbf{w}}_{k_s}^{\Delta} \\ &= +\sum_{i\neq k}^{K_s}\beta_{i_s,s}(\check{\mathbf{w}}_{k_s}^{\circ})^H \mathbf{A}_{k_s,i_s,s}\check{\mathbf{w}}_{k_s}^{\Delta}.\end{aligned} \tag{76}$$

The sufficient condition for both $(\check{\mathbf{w}}_{k_s}^{\circ})^H \mathbf{A}_{k_s,i_j,s}\check{\mathbf{w}}_{k_s}^{\Delta}$ and $(\check{\mathbf{w}}_{k_s}^{\circ})^H \mathbf{A}_{k_s,i_s,s}\check{\mathbf{w}}_{k_s}^{\Delta}$, $\forall i_j$ equaling zero can be given by $\mathcal{N}_{i_j,s}^{\mathrm{dia}} \cap \mathcal{N}_{k_s,s}^{\mathrm{dia}} = \emptyset$ or $\mathcal{N}_{i_j,s}^{\mathrm{dia}} \cap (\bar{\mathcal{N}}_{k_s,s}^{\mathrm{dia}} \setminus \mathcal{N}_{k_s,s}^{\mathrm{dia}}) = \emptyset$, $\forall i_j$, which indicates that the newly selected beam index set can be $\bar{\mathcal{N}}_{k_s,s}^{\mathrm{dia}} \setminus \mathcal{N}_{k_s,s}^{\mathrm{dia}} \subset \mathcal{N}_{i_j,s}^{\mathrm{dia}}$, if $\mathcal{N}_{i_j,s}^{\mathrm{dia}} \cap \mathcal{N}_{k_s,s}^{\mathrm{dia}} = \emptyset$, $\forall i_j$. Otherwise, $\bar{\mathcal{N}}_{k_s,s}^{\mathrm{dia}} \setminus \mathcal{N}_{k_s,s}^{\mathrm{dia}}$ should satisfy $\mathcal{N}_{i_j,s}^{\mathrm{dia}} \cap (\bar{\mathcal{N}}_{k_s,s}^{\mathrm{dia}} \setminus \mathcal{N}_{k_s,s}^{\mathrm{dia}}) = \emptyset$. Therefore, elements $\{\bar{n}_{k_s,s,m}\}_{m=1}^{\bar{N}_{k_s,s}^{\mathrm{dia}} - N_{k_s,s}^{\mathrm{dia}}}$ in the newly selected beam index set $\bar{\mathcal{N}}_{k_s,s}^{\mathrm{dia}} \setminus \mathcal{N}_{k_s,s}^{\mathrm{dia}}$ should satisfy

$$\begin{aligned}\bar{n}_{k_s,s,m} \notin \bigcup_{i\in\mathcal{K}_j, j\in\mathcal{S}, \mathcal{N}_{i_j,s}^{\mathrm{dia}} \cap \mathcal{N}_{k_s,s}^{\mathrm{dia}} \neq \emptyset} \mathcal{N}_{i_j,s}^{\mathrm{dia}}, \\ \forall s \in \mathcal{S}, k \in \mathcal{K}_s, m = 1, 2, \ldots, \bar{N}_{k_s,s}^{\mathrm{dia}} - N_{k_s,s}^{\mathrm{dia}}.\end{aligned} \tag{77}$$

It can be verified that when (77) is satisfied, $(\check{\mathbf{w}}_{k_s}^{\circ})^H \tilde{\mathbf{L}}_{k_s}\check{\mathbf{w}}_{k_s}^{\Delta}$ and $(\check{\mathbf{w}}_{k_s}^{\Delta})^H \tilde{\mathbf{L}}_{k_s}\check{\mathbf{w}}_{k_s}^{\circ}$ in (73) equal to zero. Since $\tilde{\mathbf{L}}_{k,s} \succeq \mathbf{0}$, then $(\check{\mathbf{w}}_{k_s}^{\Delta})^H \tilde{\mathbf{L}}_{k_s}\check{\mathbf{w}}_{k_s}^{\Delta} \geq 0$, which yields

$$(\check{\mathbf{w}}_{k_s}^{\diamondsuit})^H \tilde{\mathbf{L}}_{k_s}\check{\mathbf{w}}_{k_s}^{\diamondsuit} \geq (\check{\mathbf{w}}_{k_s}^{\circ})^H \tilde{\mathbf{L}}_{k_s}\check{\mathbf{w}}_{k_s}^{\circ}. \tag{78}$$

Then, we have

$$\mathrm{ASLNR}_{k_s}^{\mathrm{b}}\left(\check{\mathbf{w}}_{k_s}^{\mathrm{aslnr}}, \bar{\mathcal{N}}_{k_s,s}^{\mathrm{dia}}\right) \leq \mathrm{ASLNR}_{k_s}^{\mathrm{b}}\left(\check{\mathbf{w}}_{k_s}^{\mathrm{aslnr}}, \mathcal{N}_{k_s,s}^{\mathrm{dia}}\right). \tag{79}$$

According to (68) and (79), we can obtain that

$$\mathrm{ASLNR}_{k_s}^{\mathrm{b}}\left(\check{\mathbf{w}}_{k_s}^{\mathrm{aslnr}}, \bar{\mathcal{N}}_{k_s,s}^{\mathrm{dia}}\right) = \mathrm{ASLNR}_{k_s}^{\mathrm{b}}\left(\check{\mathbf{w}}_{k_s}^{\mathrm{aslnr}}, \mathcal{N}_{k_s,s}^{\mathrm{dia}}\right). \tag{80}$$

This completes the proof.

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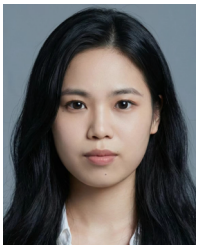
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